COS20019 - Cloud Computing Architecture

Assignment 1 - Part B
Creating and deploying Photo Album website onto a simple AWS infrastructure

Author: Trac Duc Anh Luong - ID: 103488117

Due Date: 15/10/2023

Name: Trac Duc Anh Luong - ID: 103488117

Assignment 1b rubric		
Criteria	**Page number**	**Pts**
Infrastructure Requirements		
VPC with 2 public and 2 private subnets	2-3	0.5 pts
Correct Public and Private Routing tables with correct subnet associations	4	1 pts
Security groups properly configured and attached.	5-7	1 pts
Network ACL properly configured and attached	8-10	1.5 pts
Correct Web server and Test instances running in correct subnets	11-15	0.5 pts
Database schema as specified	16-20	0.5 pts
Database running in correct subnets	21-22	1 pts
S3 objects publicly accessible, using proper access policy	22-23	0.5 pts
Functional Requirements		
album.php page displayed from EC2 Web server	24-25	1 pts
Provided URL is persistent (Elastic IP Association)	25-26	0.5 pts
Photos loaded from S3 with matching metadata from RDS	27-28	1 pts
Web server instance reachable from Test instance via ICMP	29	1 pts
Deductions		
Documentation not as specified or poorly presented (up to minus 20)	None	0 pts
Serious misconfigurations of AWS services being used (up to minus 20)	None	0 pts
		Total points: 10

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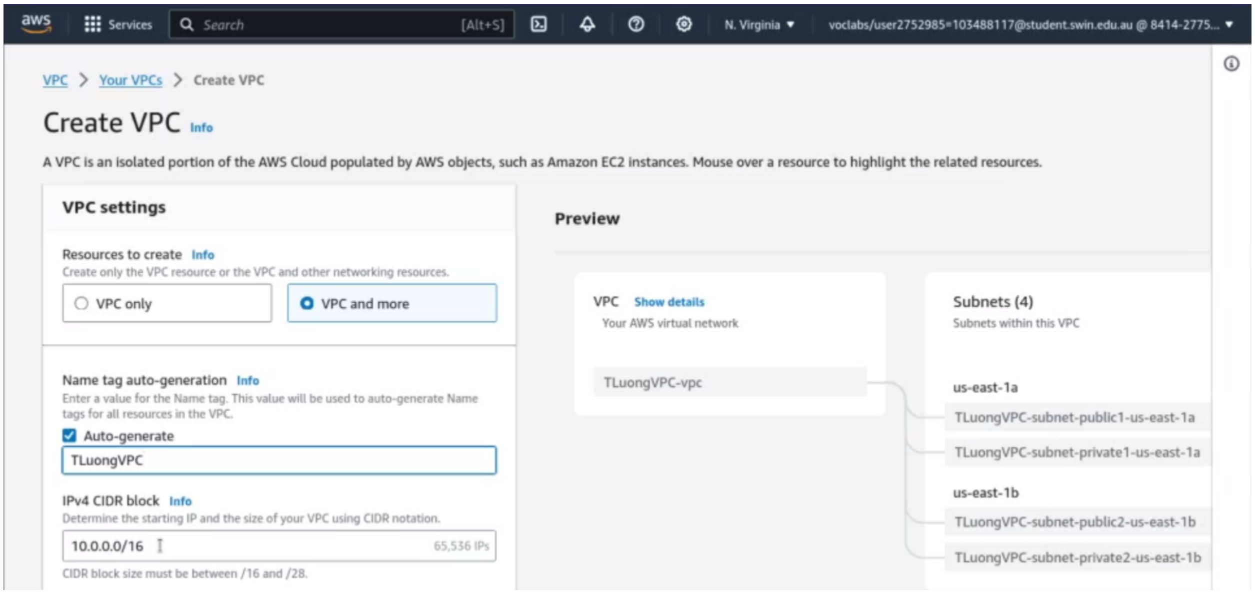
Infrastructure Requirements

VPC with 2 public and 2 private subnets

1. Create a new VPC
   - Name: TLuongVPC
   - Region: us-east-1
   - IPv4 CIDR block: 10.0.0.0/16



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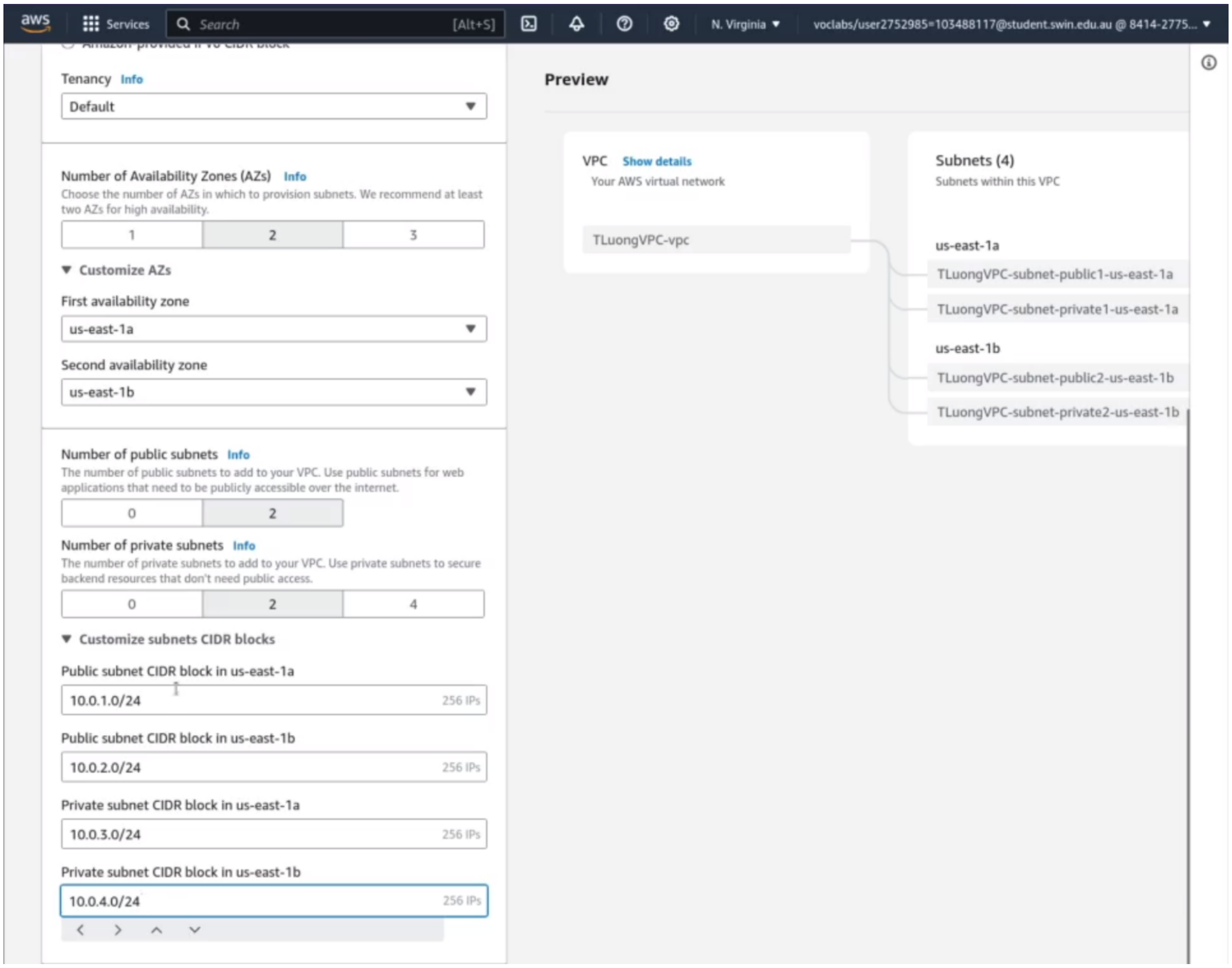
2. Number of AZs
    - First availability zone: us-east-1a
    - Second availability zone: us-east-1b
3. Number of subnets
    - Public: 2
    - Private: 2
    - Customise subnets CIDR blocks: Follow the diagram in the assignment specification.



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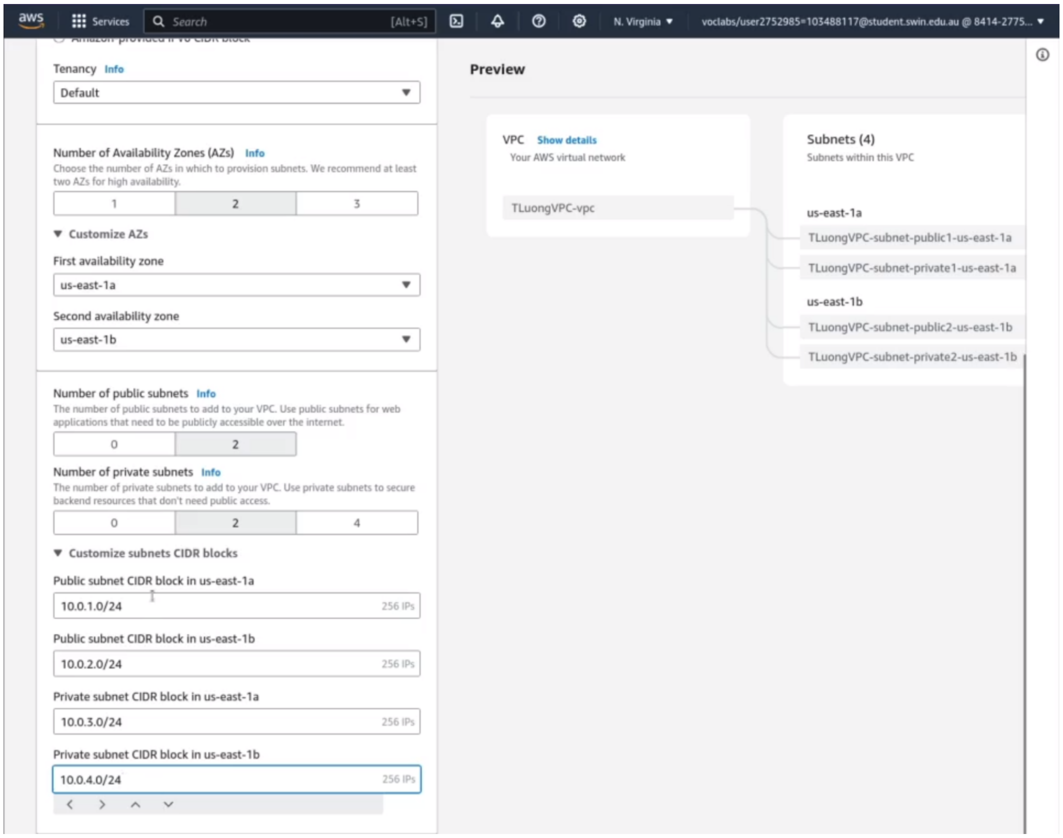
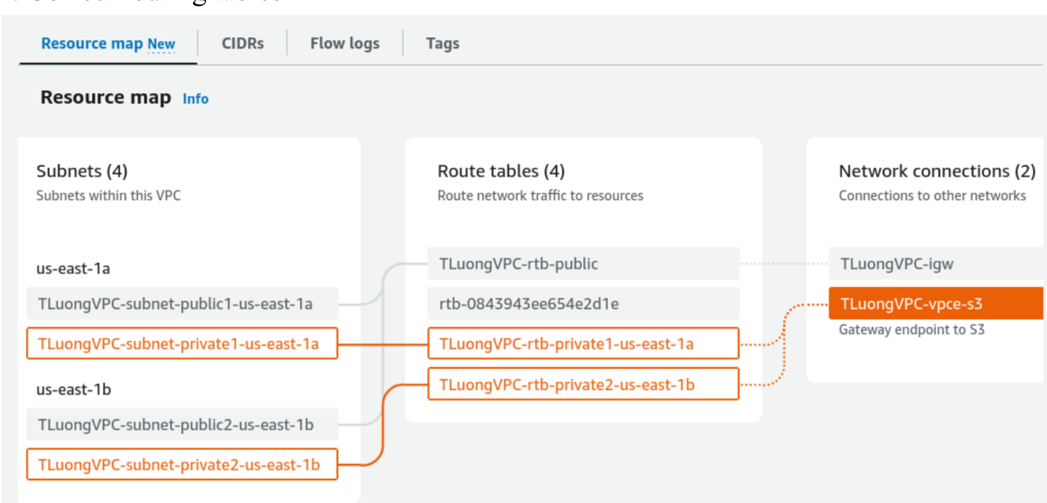
Correct Public and Private Routing tables with correct subnet associations

1. Correct subnet associations



2. Correct routing tables



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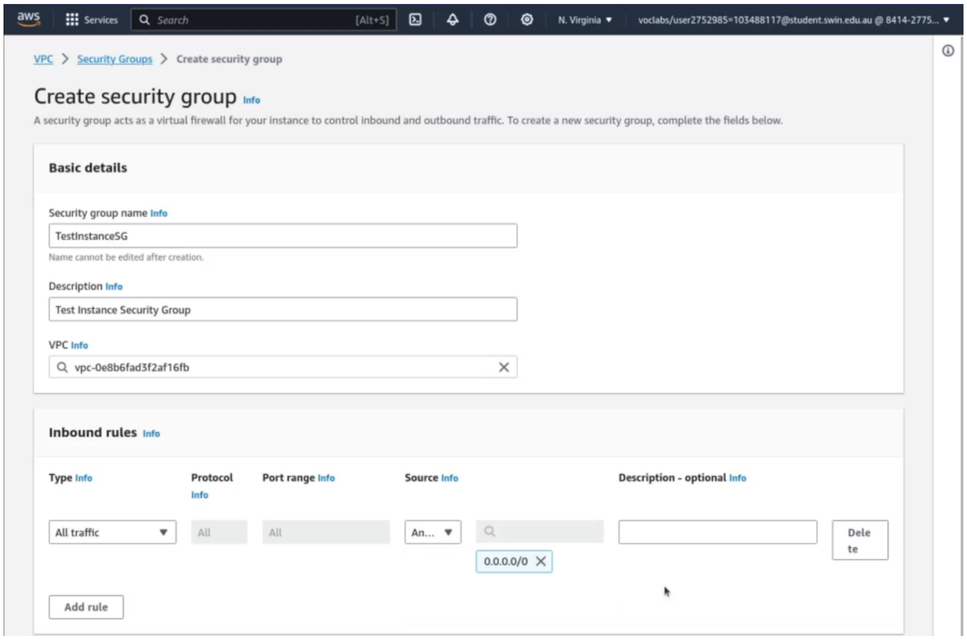
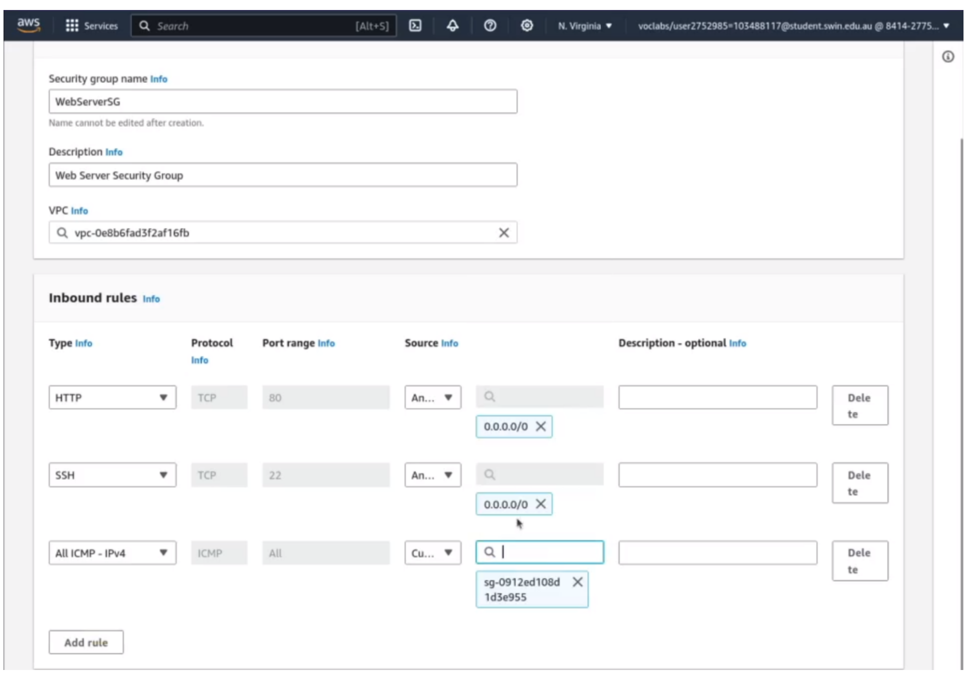
Security groups properly configured and attached.

1. Security groups configured
    a. TestInstanceSG: All traffic - Anywhere



    b. WebServerSG:
        i. HTTP (80) - Anywhere
        ii. SSH (22) - Anywhere
        iii. All ICMP - IPv4 - TestInstanceSG



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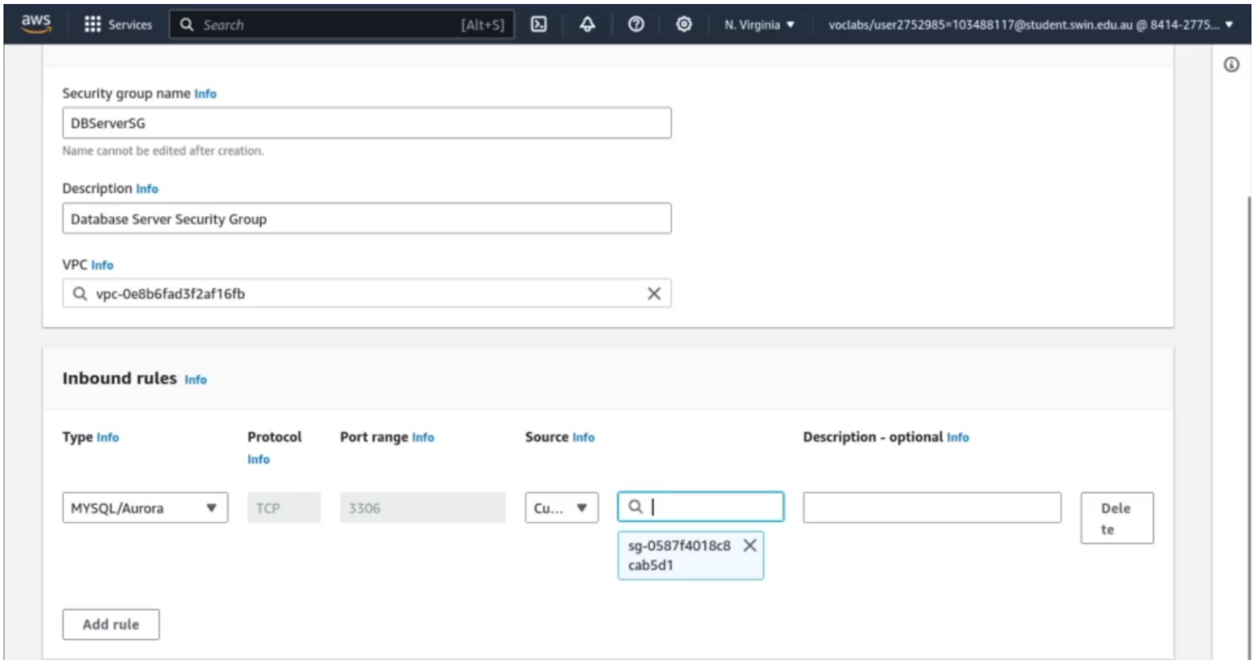
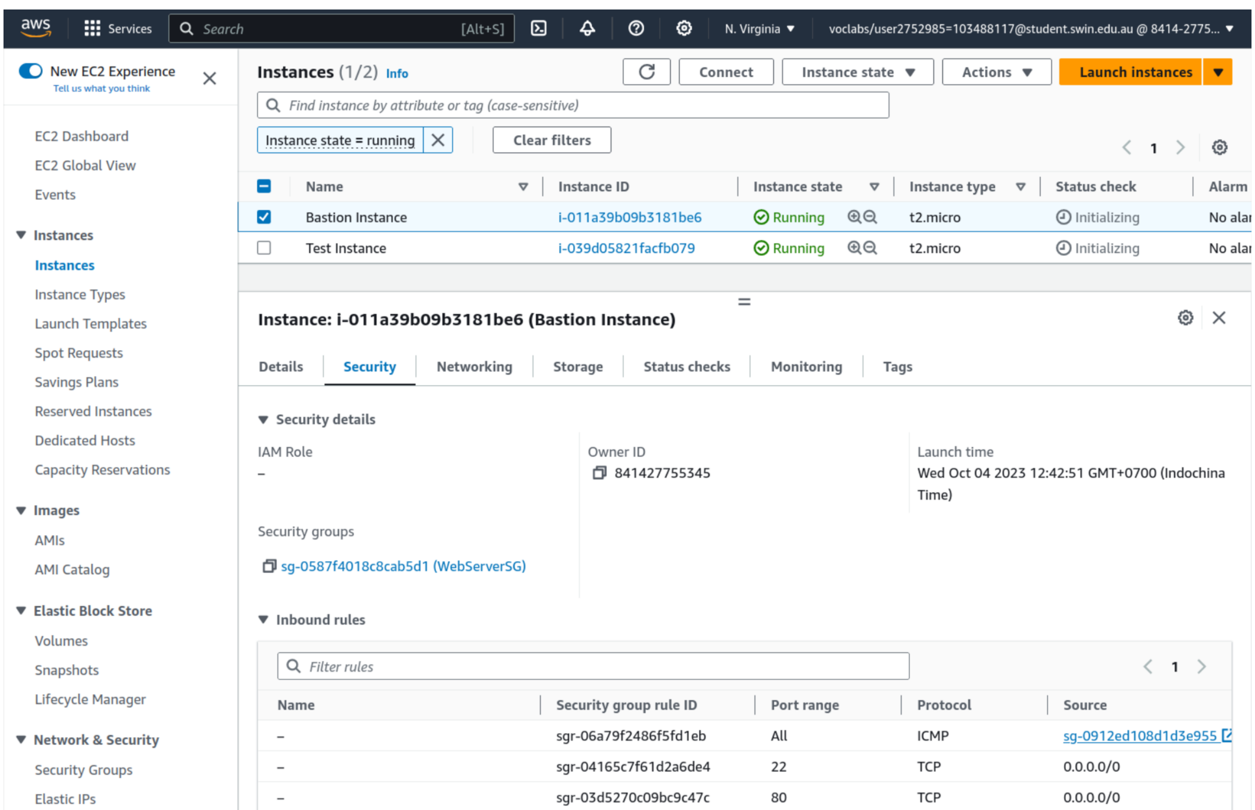
c. DBServerSG

i. MySQL (3306) - WebServerSG



2. Security groups attached

a. WebServerSG attached to Bastion Instance EC2



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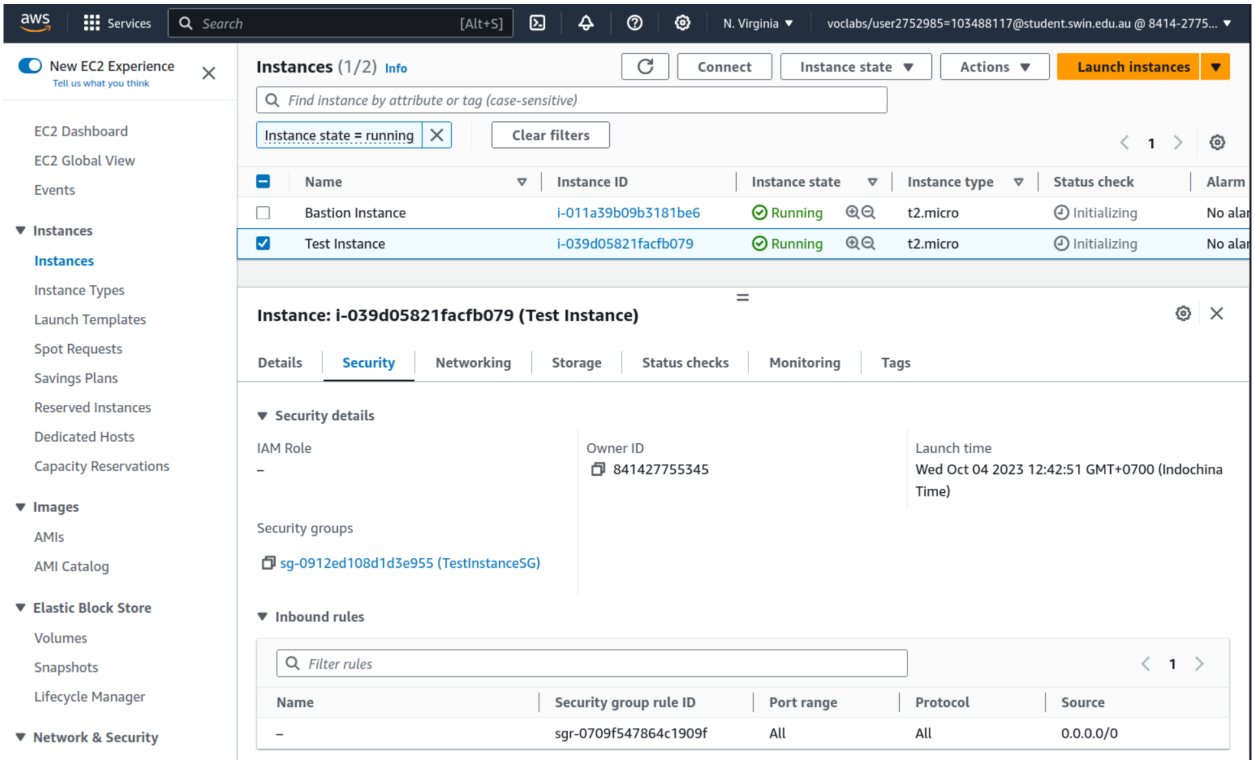
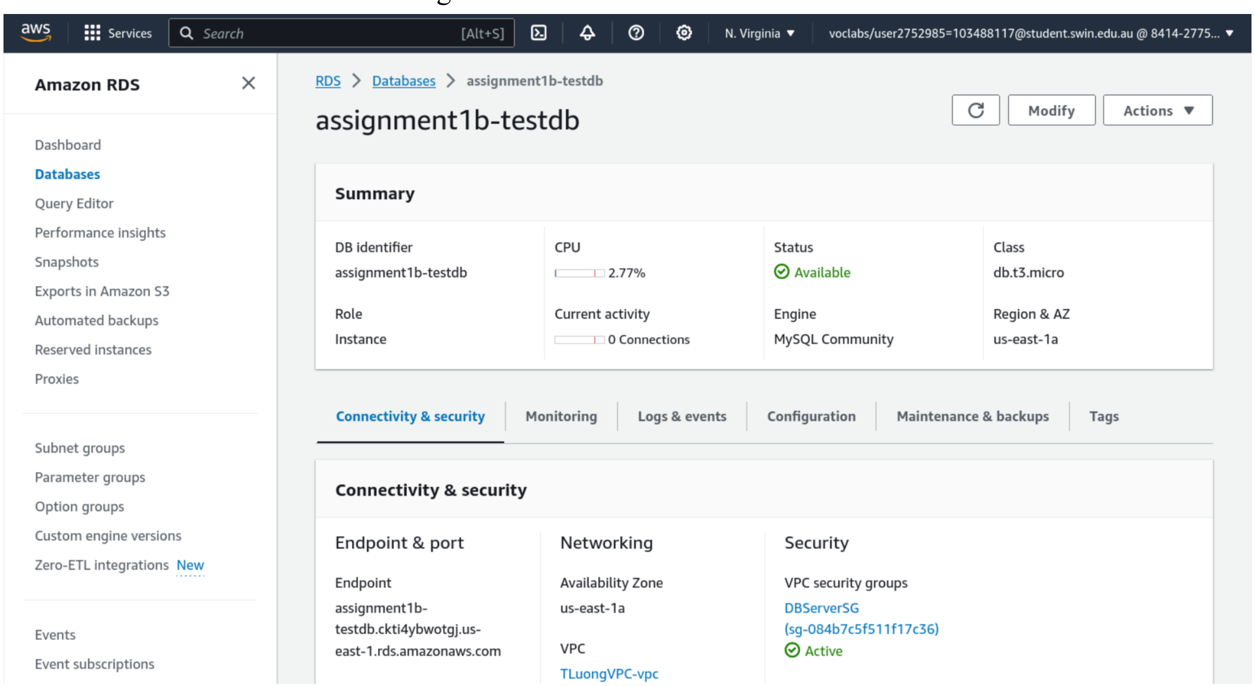
b. TestInstanceSG attached to Test Instance EC2



c. DBServerSG attached to assignment1b-testdb RDS



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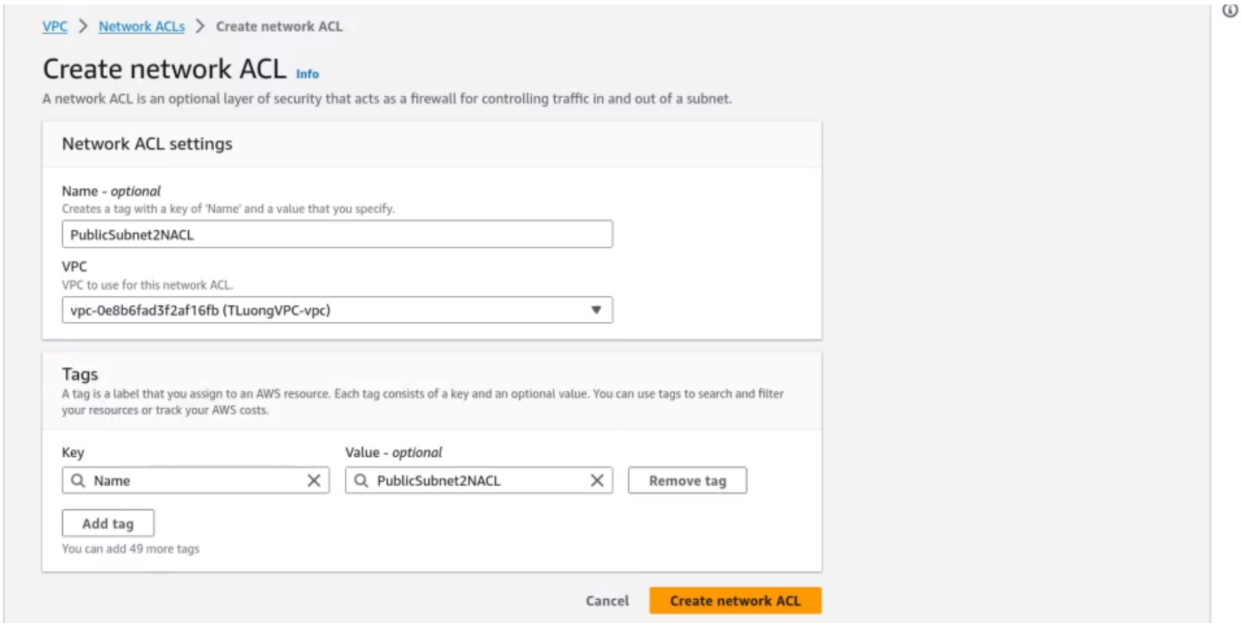
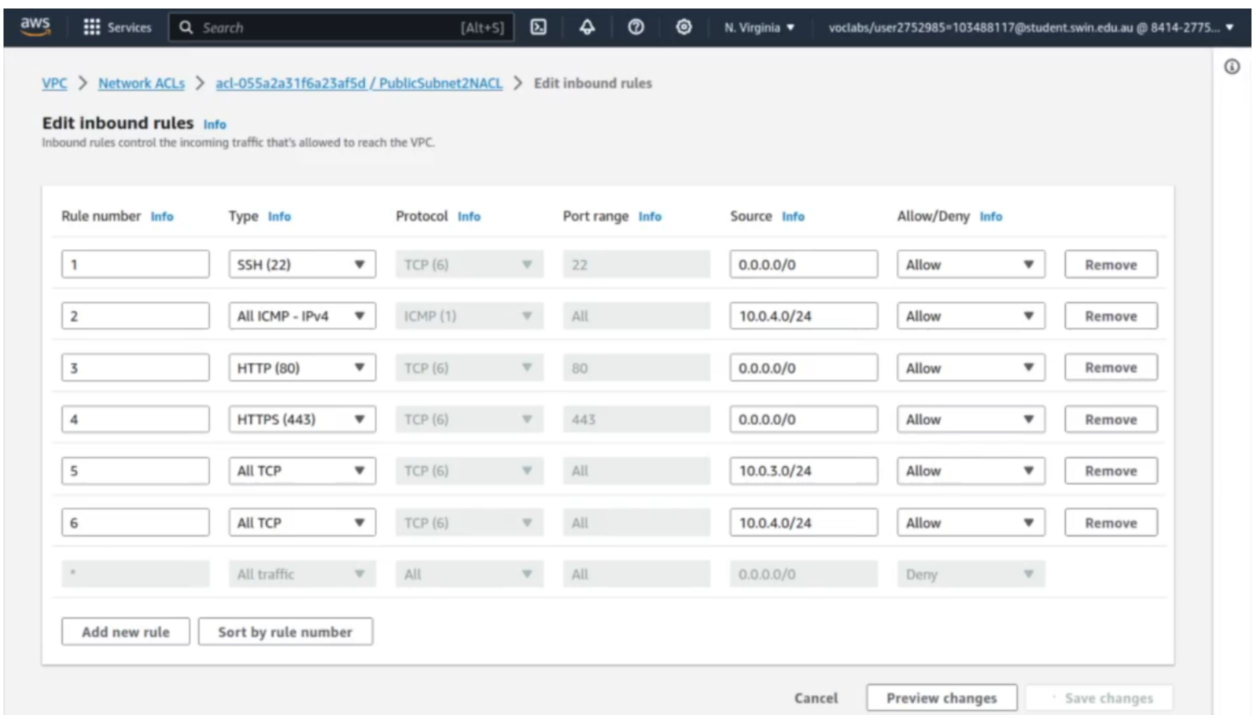
Network ACL properly configured and attached

1. Create new network ACL: PublicSubnet2NACL



2. Edit inbound rules



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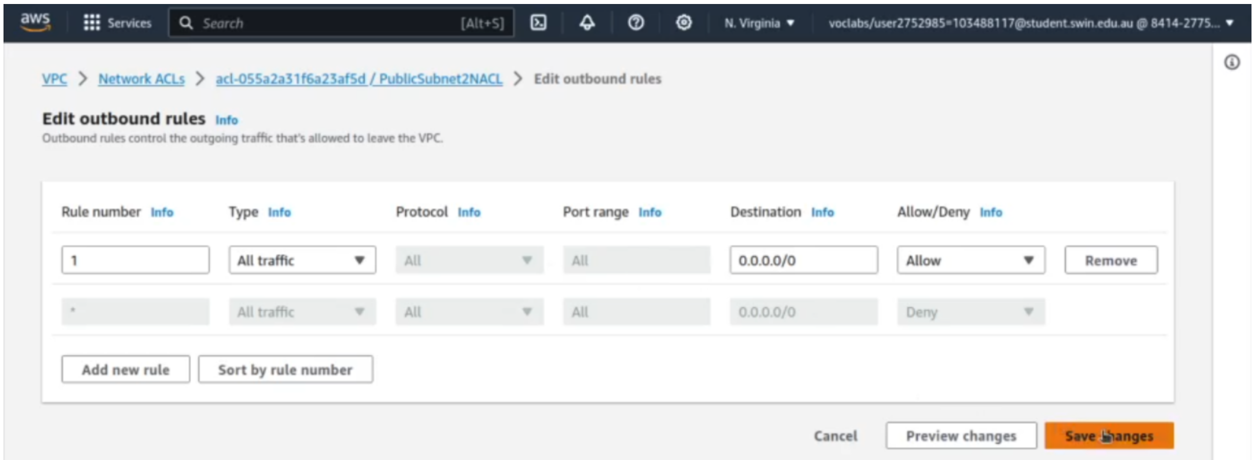
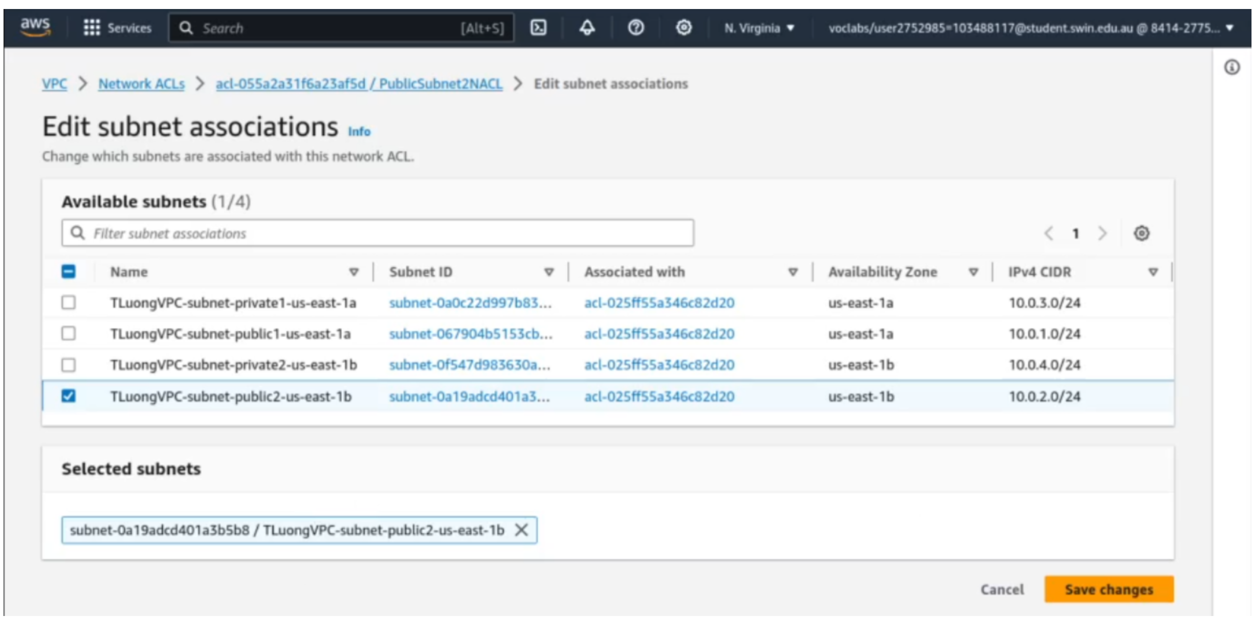
3. Edit outbound rules



4. Edit subnet association

Choose Public Subnet 2

- AZ: us-east-1b
- IPv4CIDR: 10.0.2.0/24



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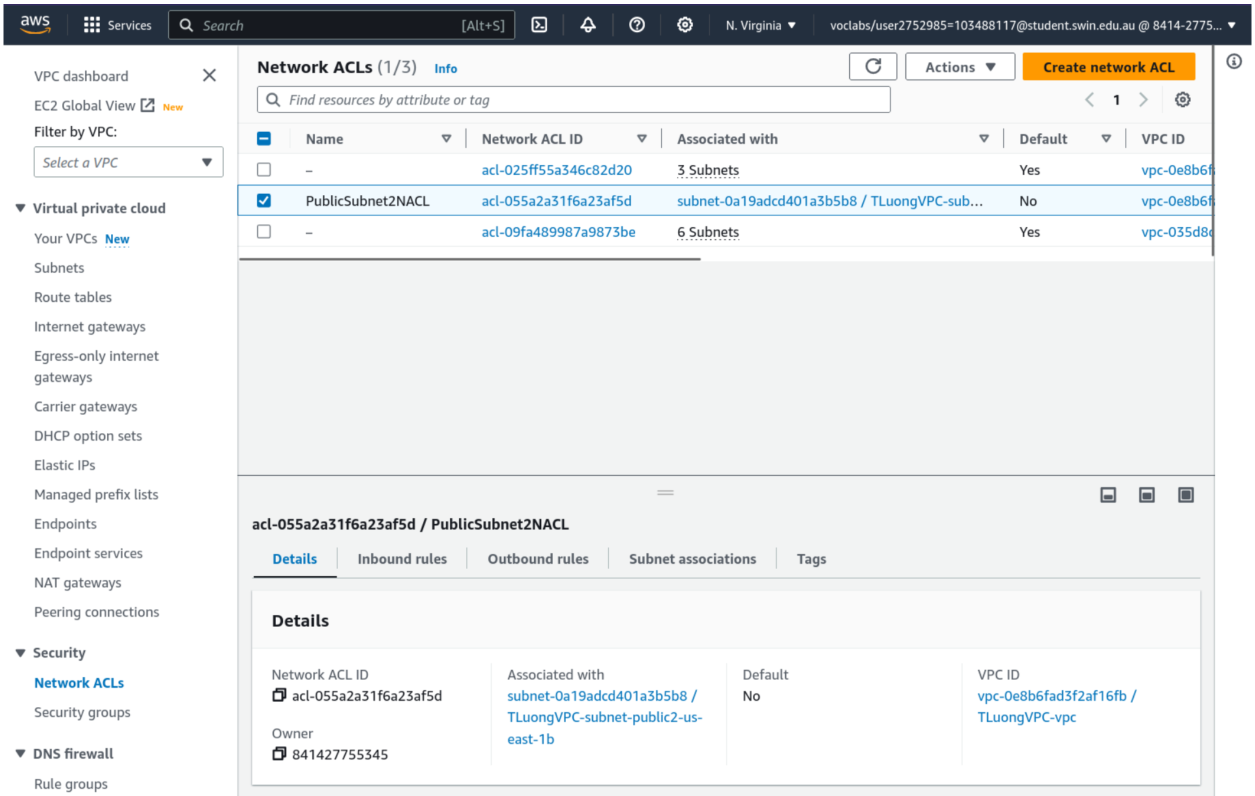
Confirm association



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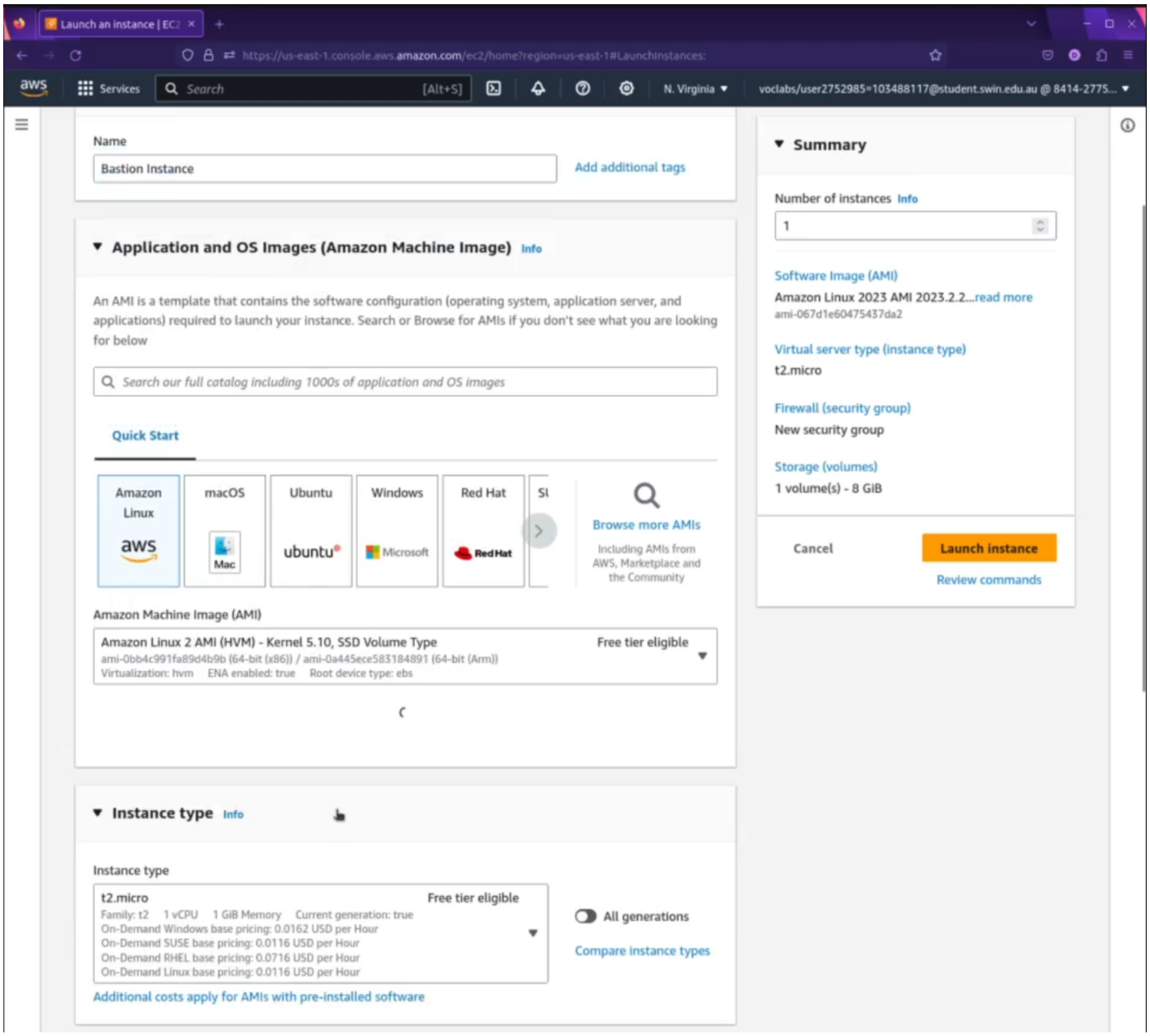
Correct Web server and Test instances running in correct subnets

1. Create a web server instance
   - Name: Bastion Instance
   - AMI: Amazon Linux 2 AMI (HVM), SSD Volume Type
   - Instance type: t2.micro



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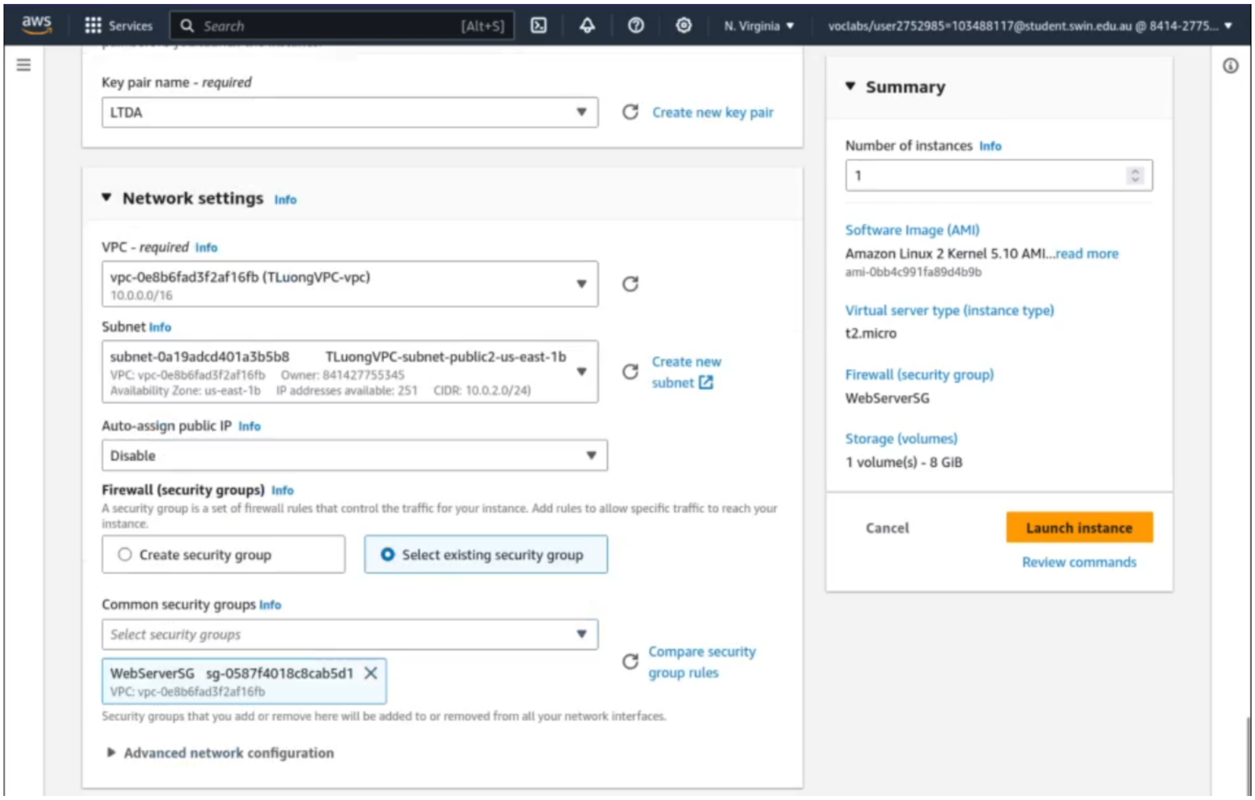
- VPC: TLuongVPC
- Subnet: TLuongVPC-subnet-public2-us-east-1b
- Security group: WebServerSG



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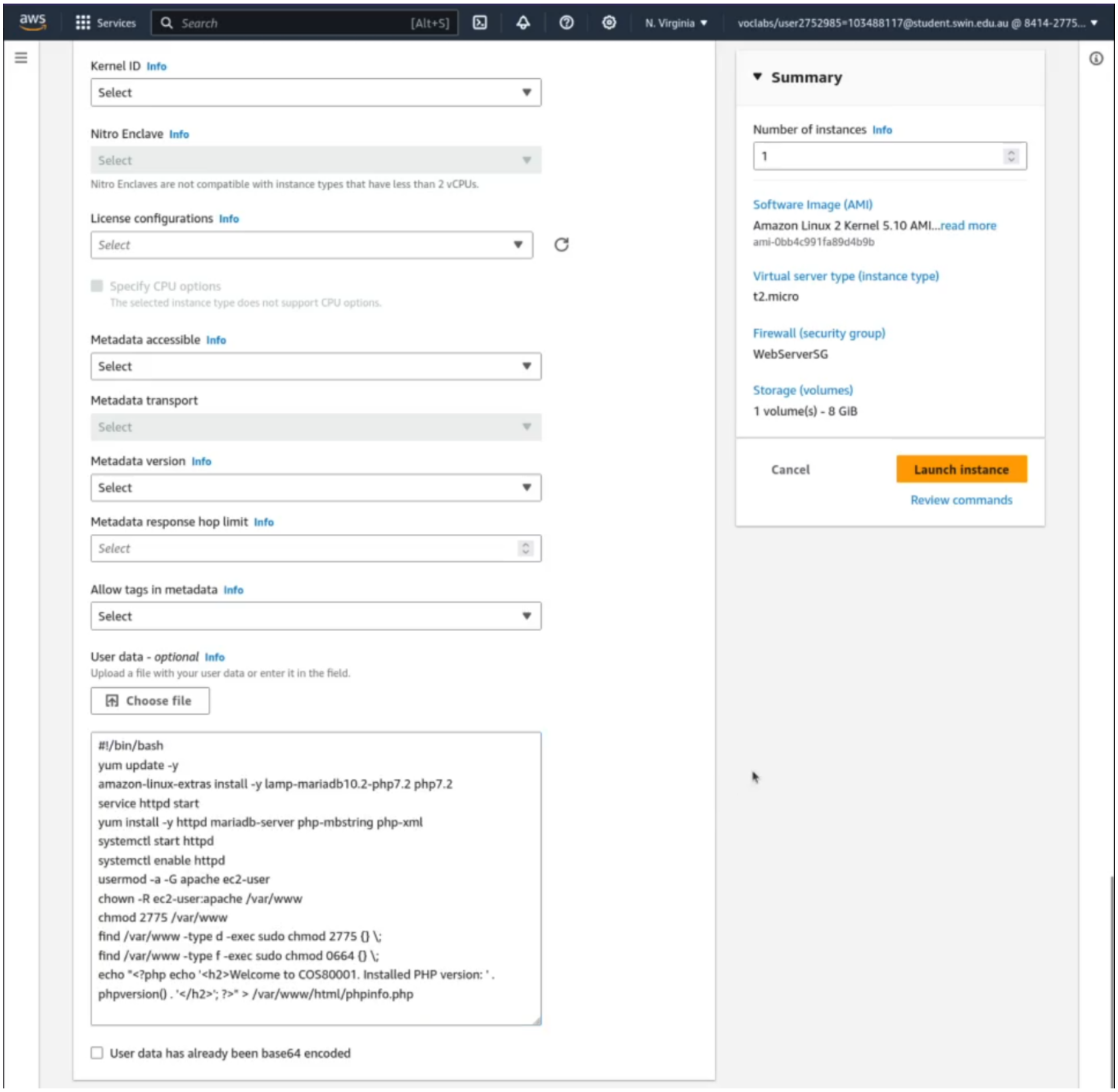
- Advanced details: paste the bash script from assignment 1a to the user data text area



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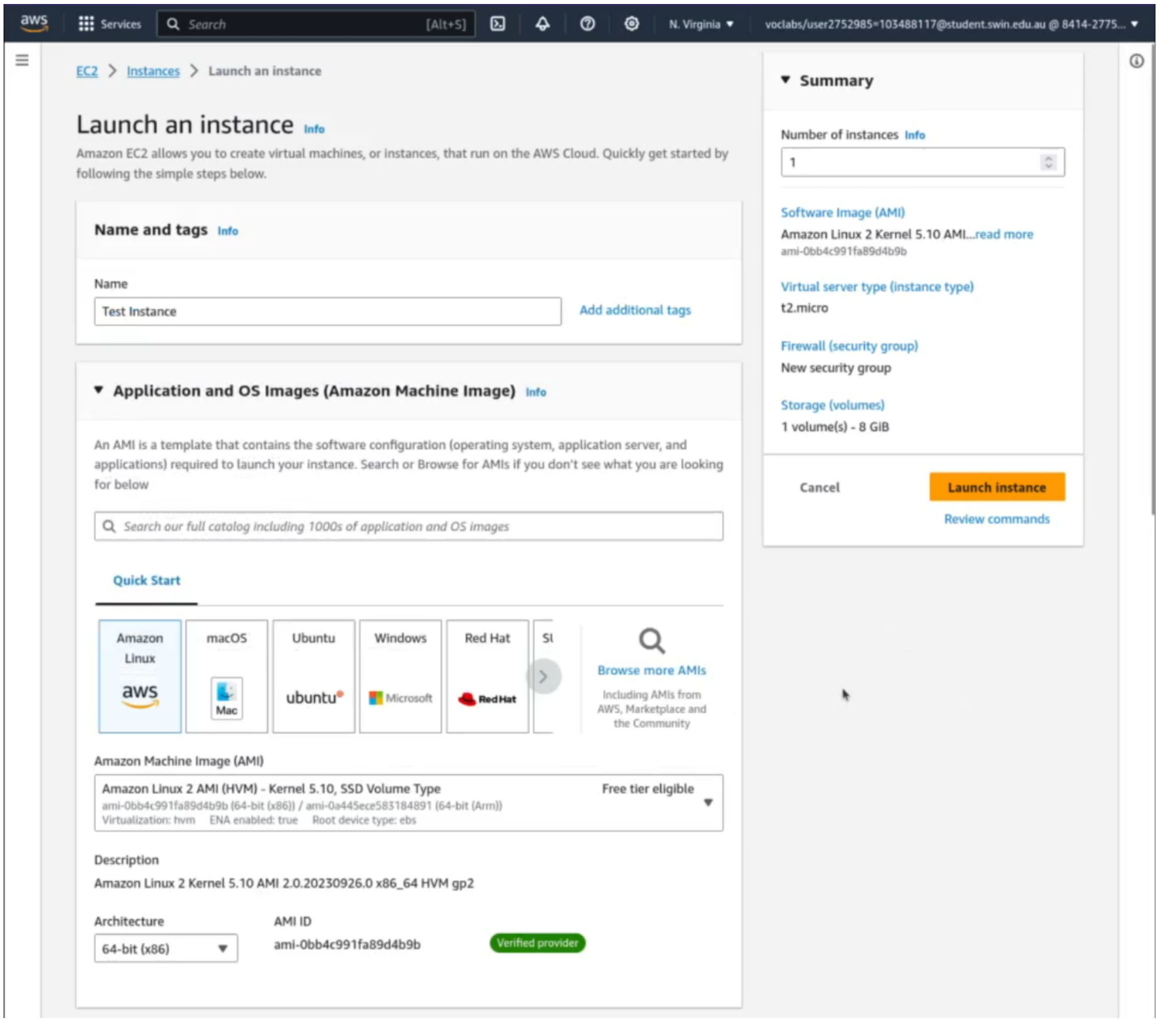
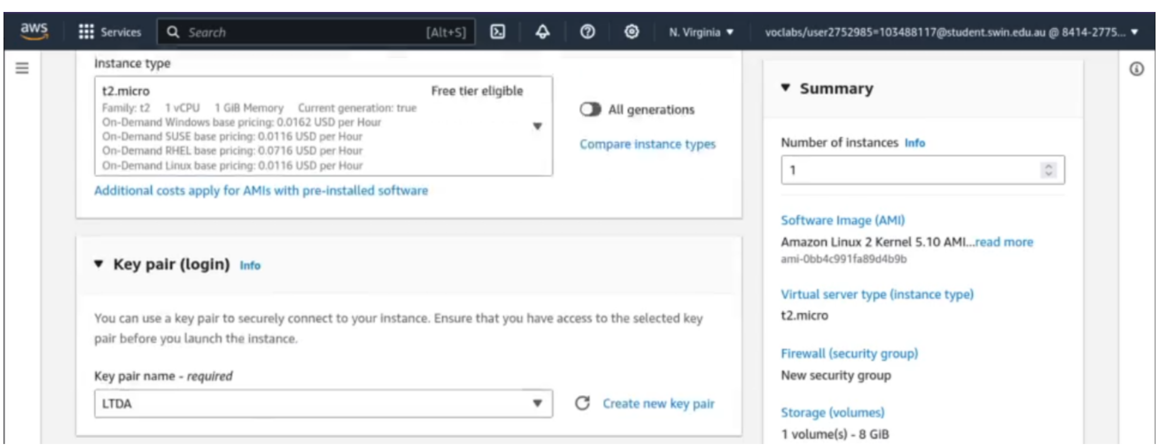
2. Create a test instance
   - Name: Bastion Instance
   - AMI: Amazon Linux 2 AMI (HVM), SSD Volume Type
   - Instance type: t2.micro





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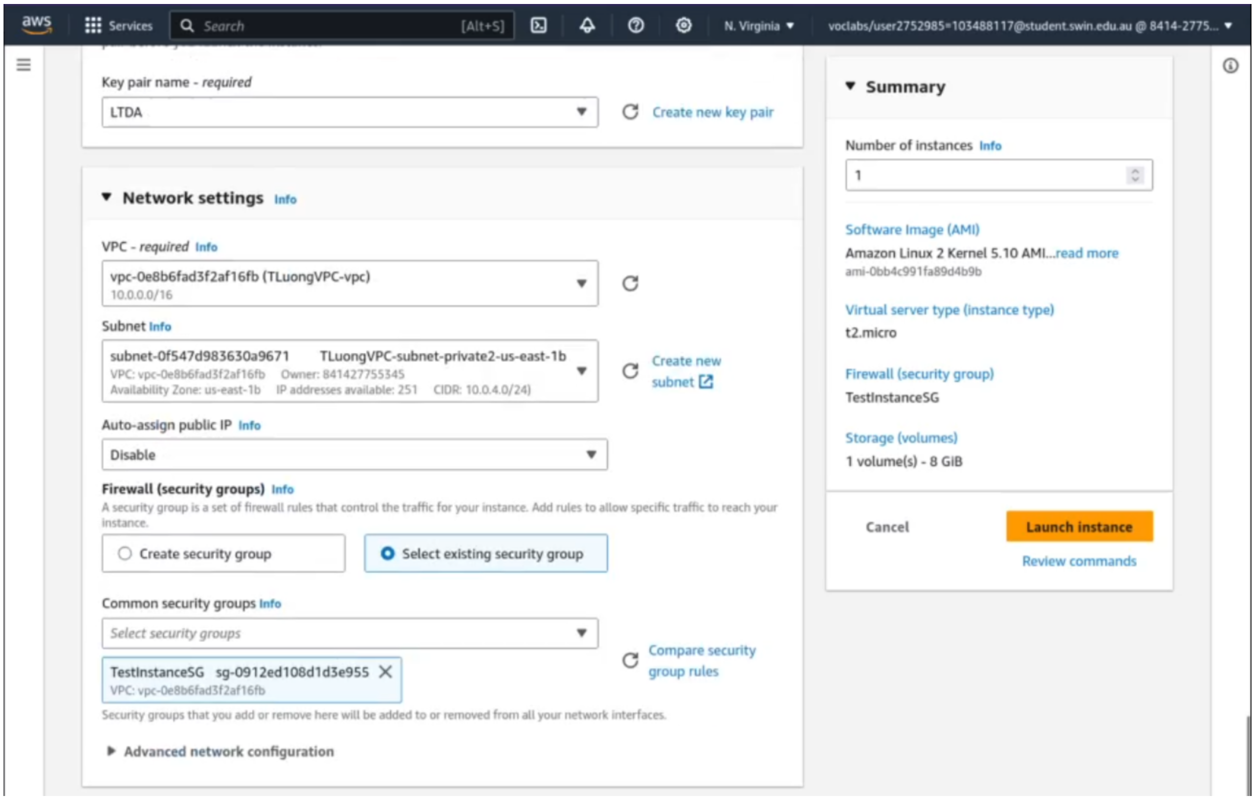
- VPC: TLuongVPC
- Subnet: TLuongVPC-subnet-private2-us-east-1b
- Security group: TestInstanceSG



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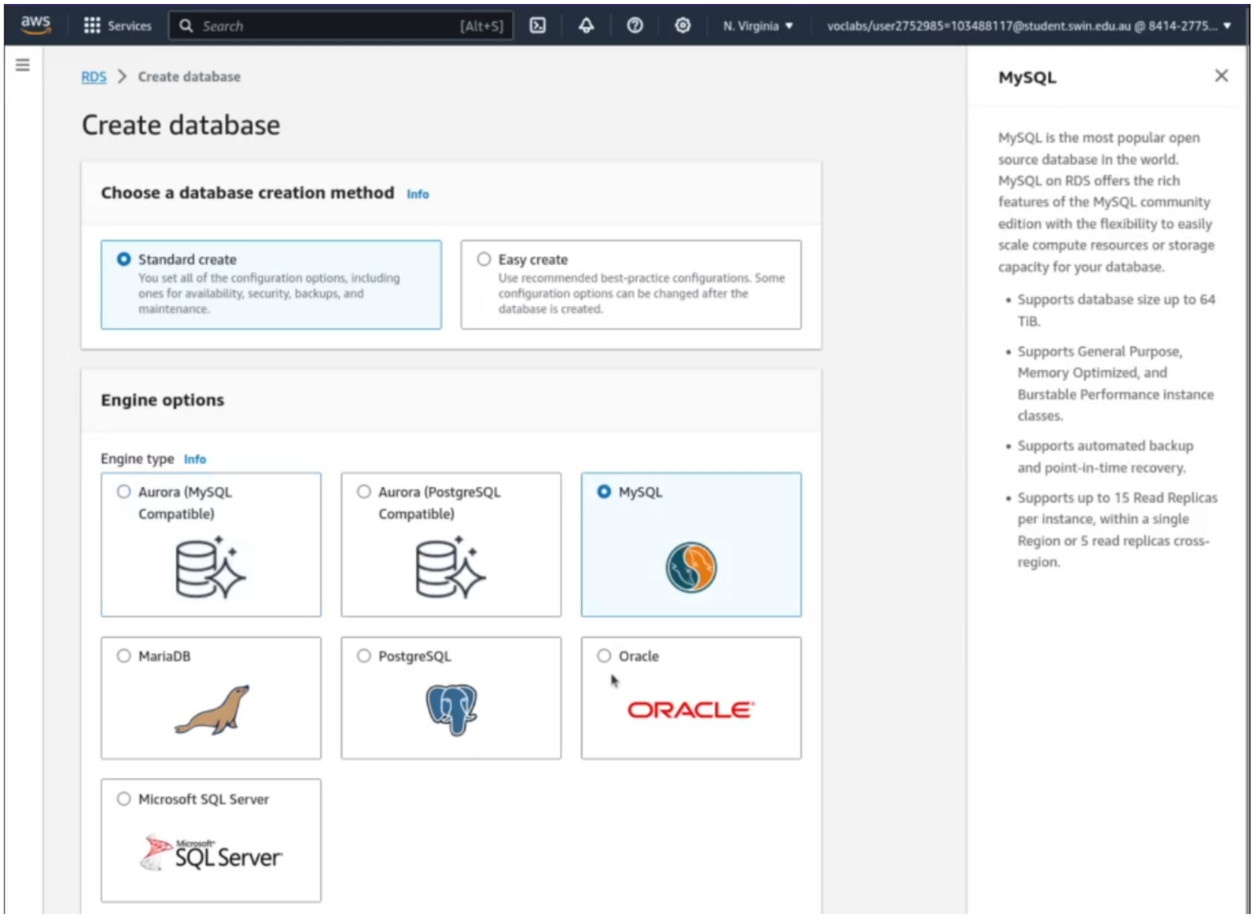
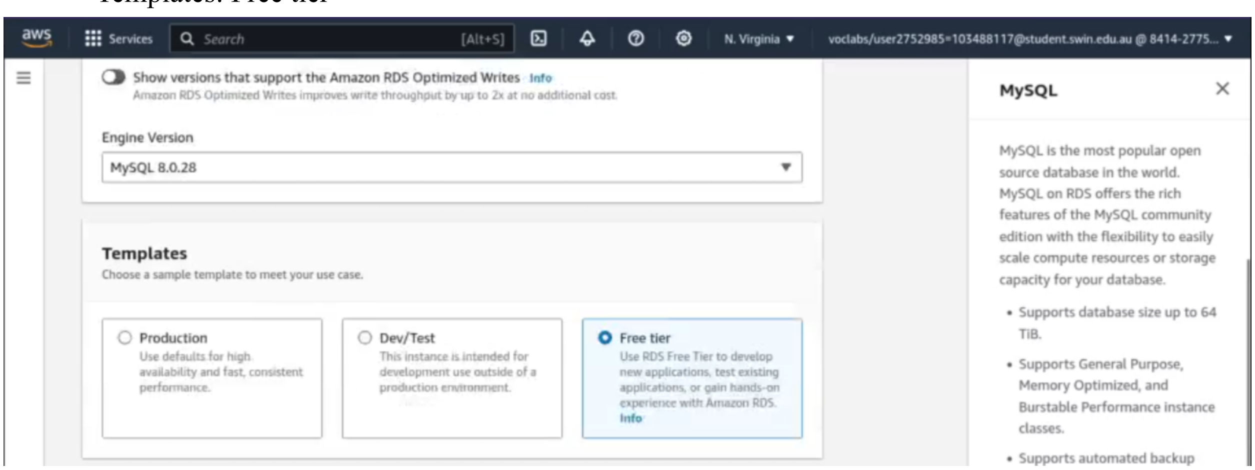
Database schema as specified

1. Create database

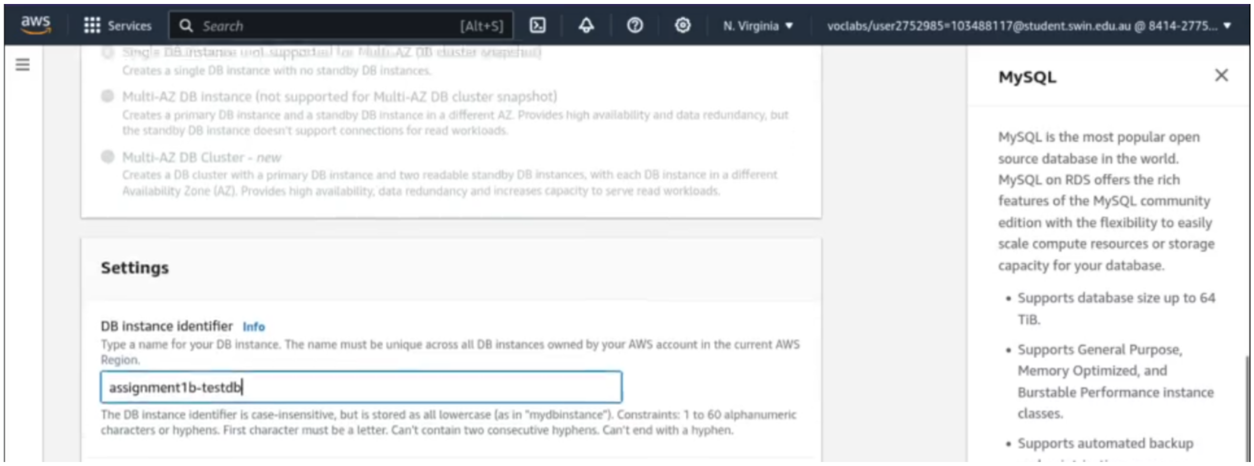
- Creation method: standard create
- Engine type: MySQL



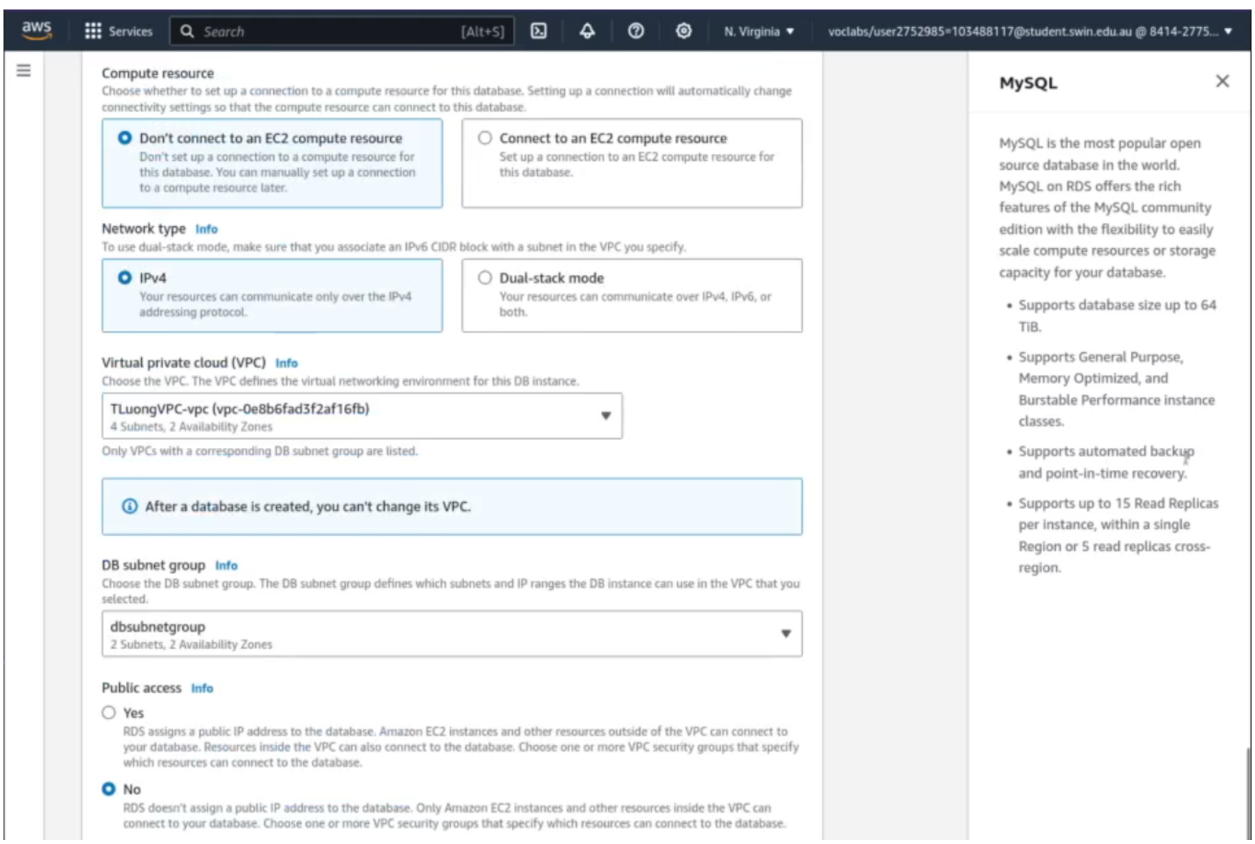
- Engine version: 8.0.28 (there was no 8.0.25 option like in the assignment specification)
- Templates: Free tier



- DB instance identifier: assignment1b-testdb



- Compute resource: Don’t connect to an EC2 compute resource (Do this later)
- Network type: IPv4
- VPC: TLuongVPC
- DB subnet group: dbsubnetgroup (create this before creating the database, I will provide the details in the next requirement
- Public access: No



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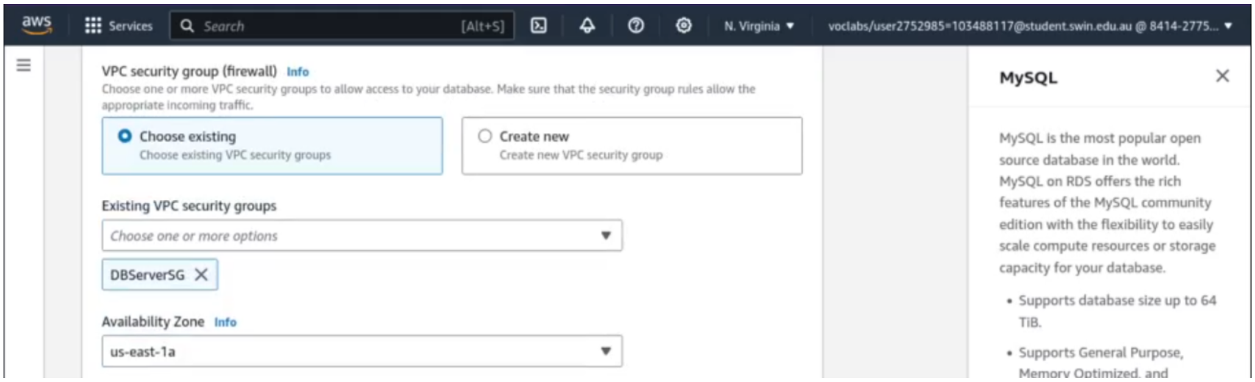
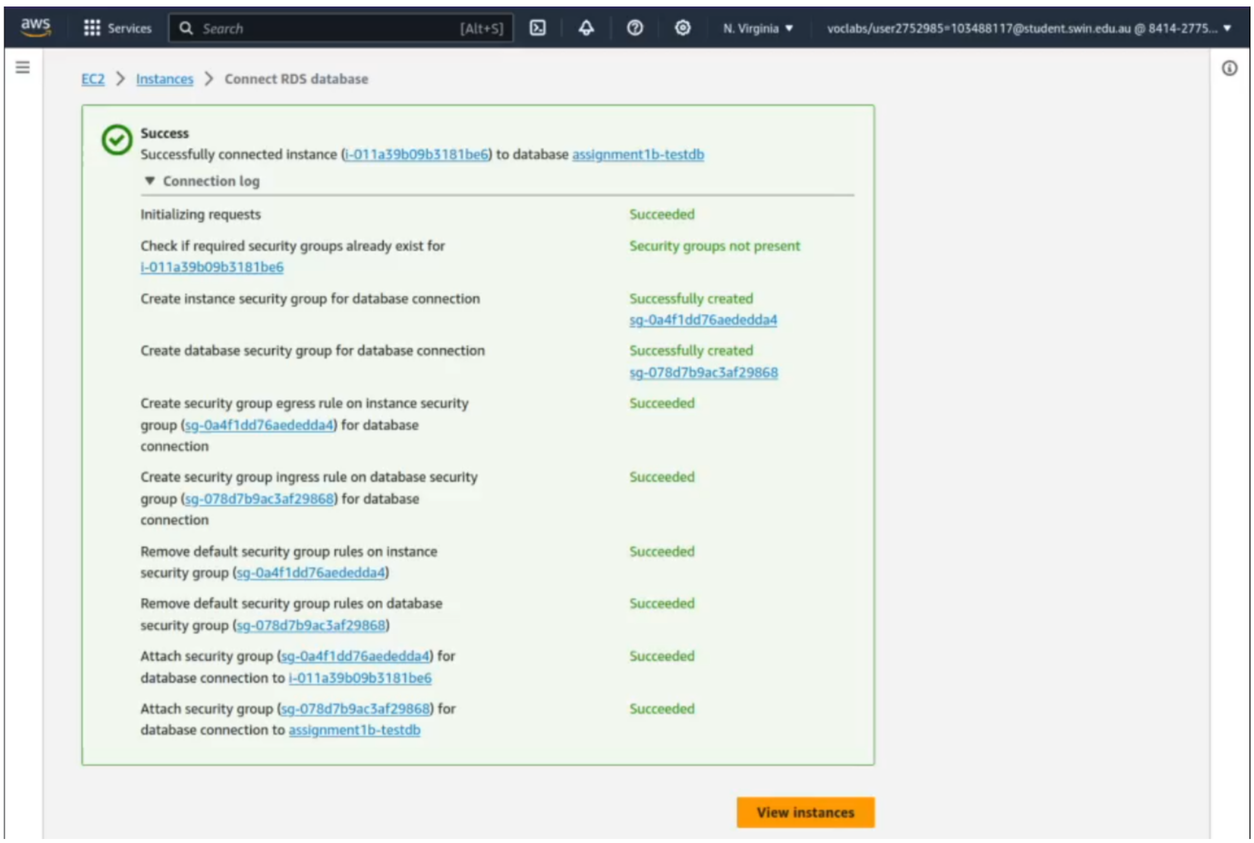
- VPC security group: DBServerSG
- AZ: us-east-1a



2. Connect EC2 Bastion Instance to the created RDS database



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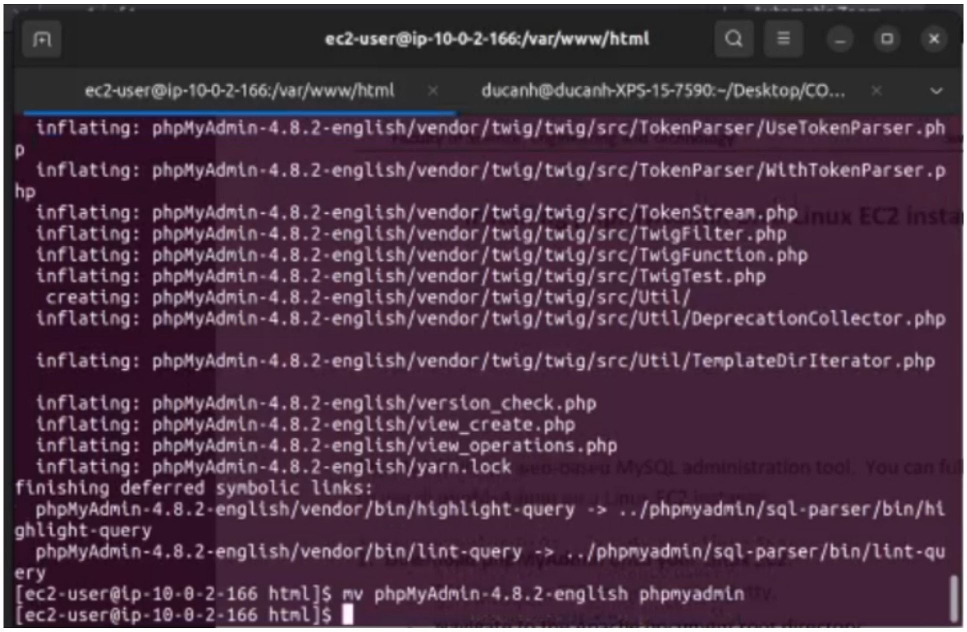
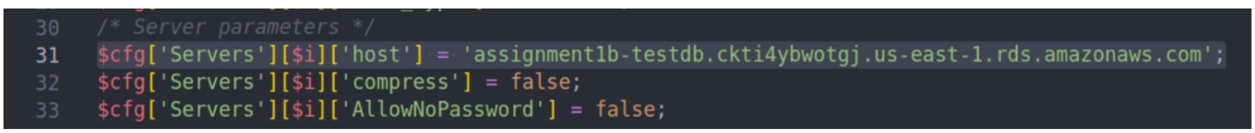
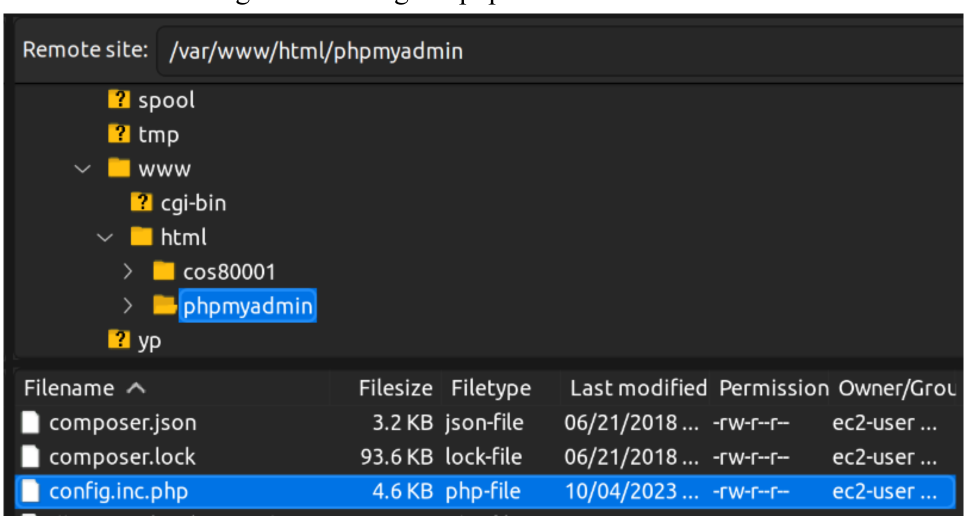
3. Install PhpMyAdmin in the EC2 Bastion Instance



4. Edit the config.sample.inc.php file and change the host from localhost to the RDS endpoint



5. Rename the config file to config.inc.php



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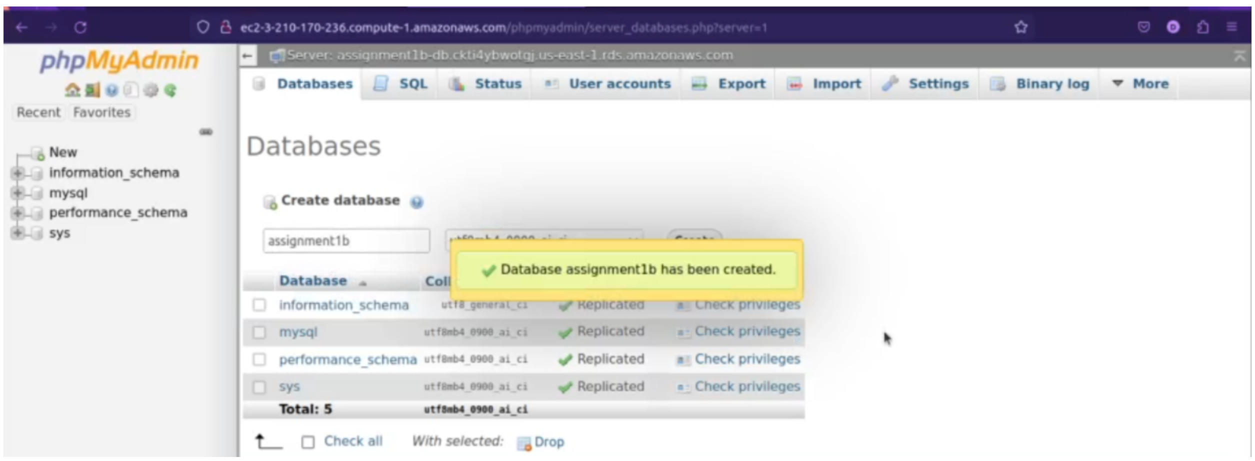
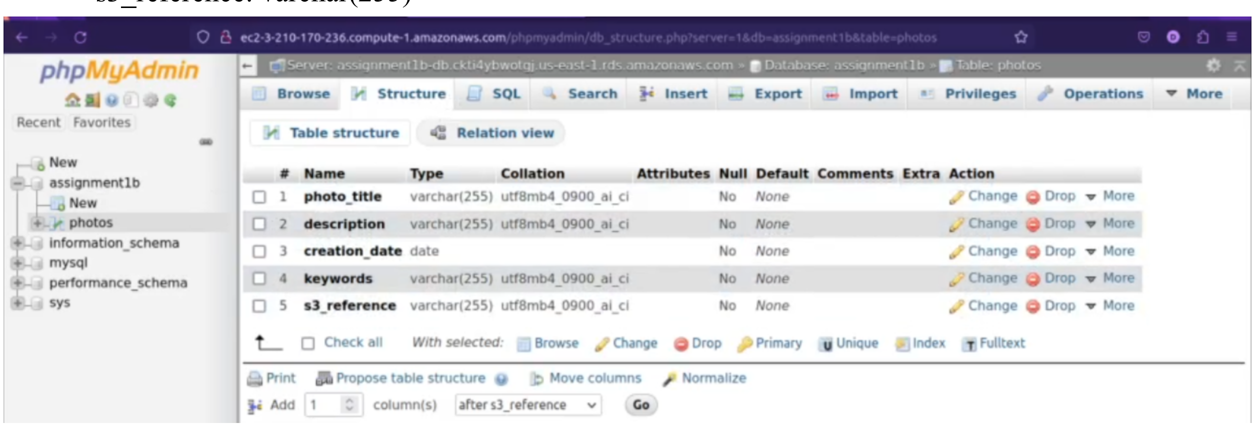
6. Login to PhpMyAdmin using the public dns + /phpmyadmin/ link, the credential has already been defined in the RDS creation step.

7. Create database assignment1b



8. Create table photos with the required columns:

- photo_title: varchar(255)
- description: varchar(255)
- creattion_date: date
- keywords: varchar(255)
- s3_reference: varchar(255)



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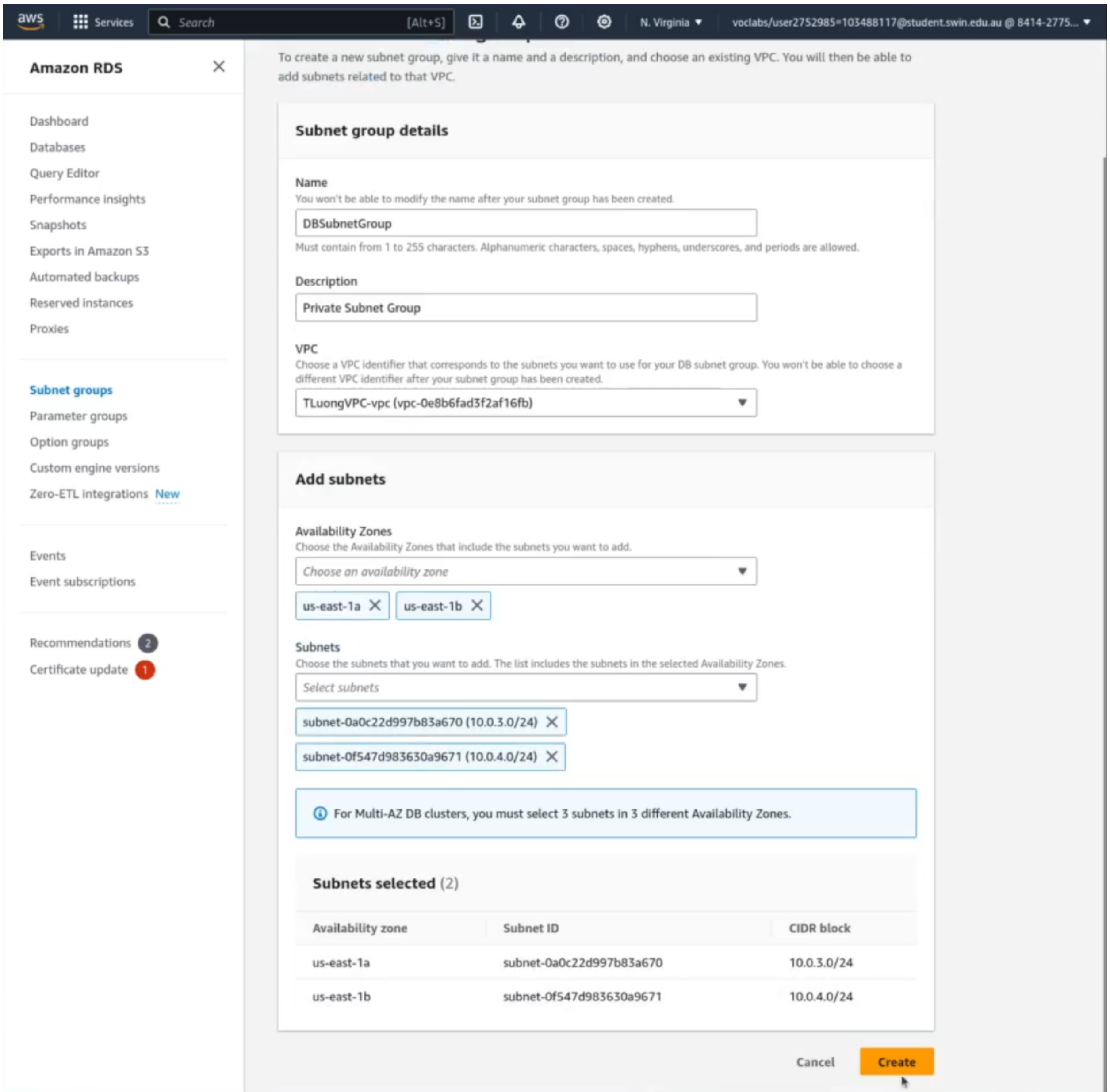
Database running in correct subnets

1. Create RDS Subnet group
   - Name: DBSubnetGroup
   - VPC: TLuongVPC
   - AZs: us-east-1a, us-east-1b
   - Subnets: private subnets (10.0.3.0/24 and 10.0.4.0/24)



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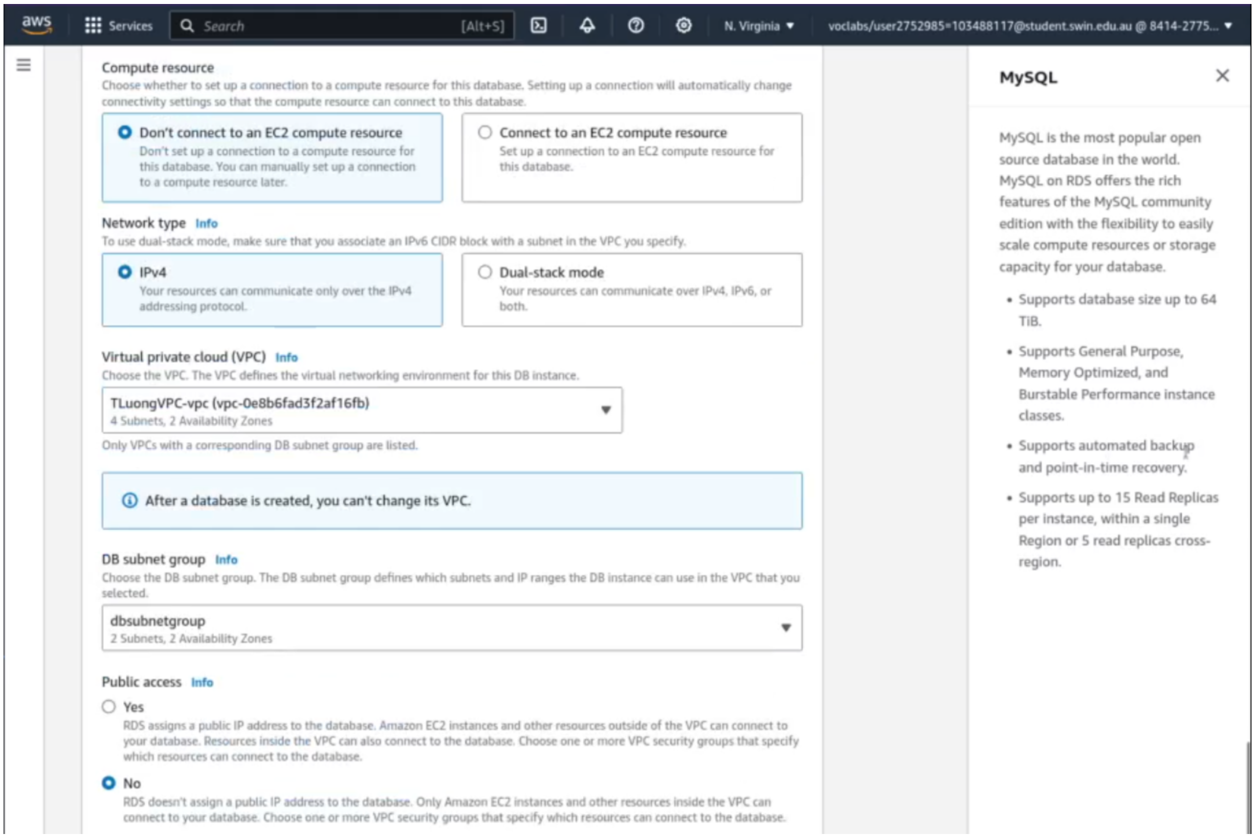
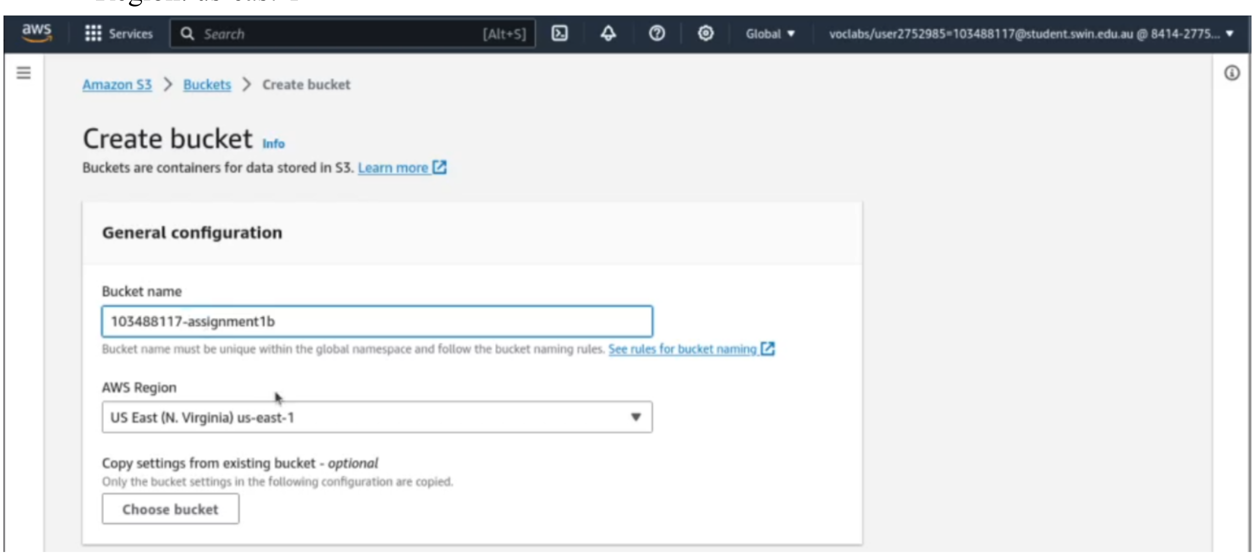
2. Connect the RDS database to the subnet group (Done in the RDS creation step)



S3 objects publicly accessible, using proper access policy

1. Create bucket
   - Name: 103488117-assignment1b
   - Region: us-east-1



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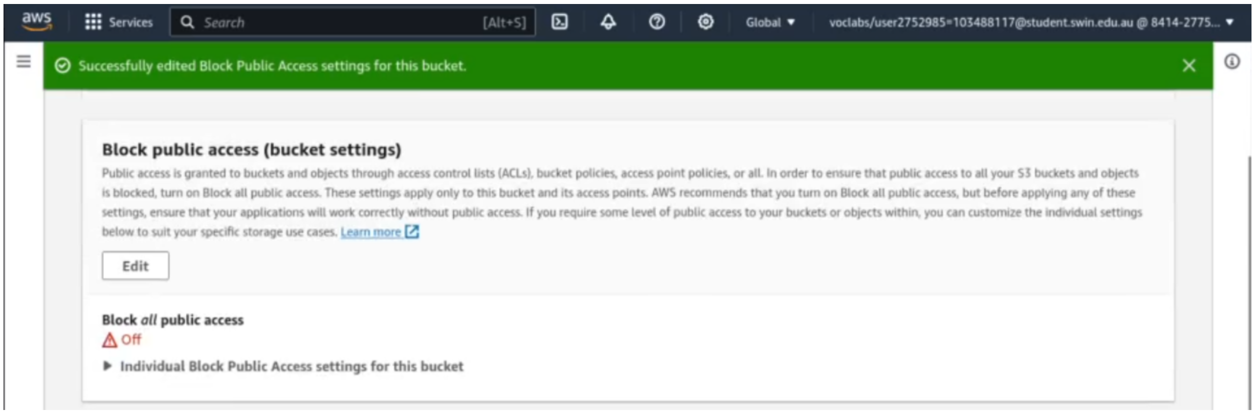
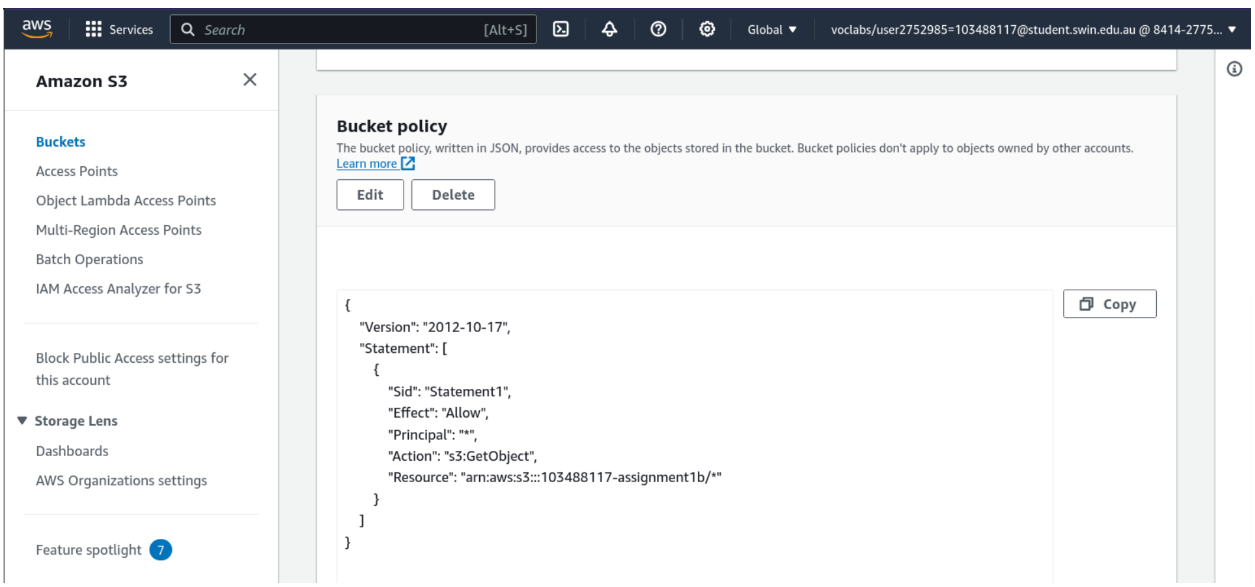
2. Turn off block all public access



3. Edit bucket policy

- Effect: Allow
- Principal: *
- Action: s3:GetObject
- Resource: [Bucket_ARN]/* (all resources)



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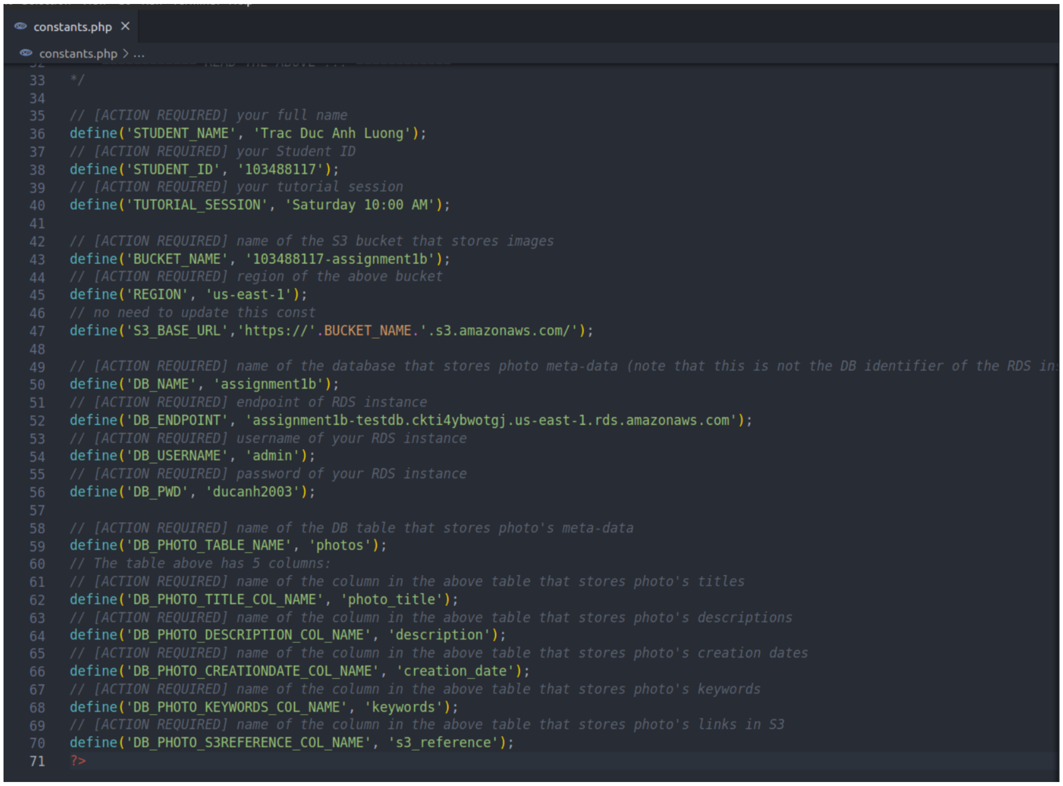
Functional Requirements

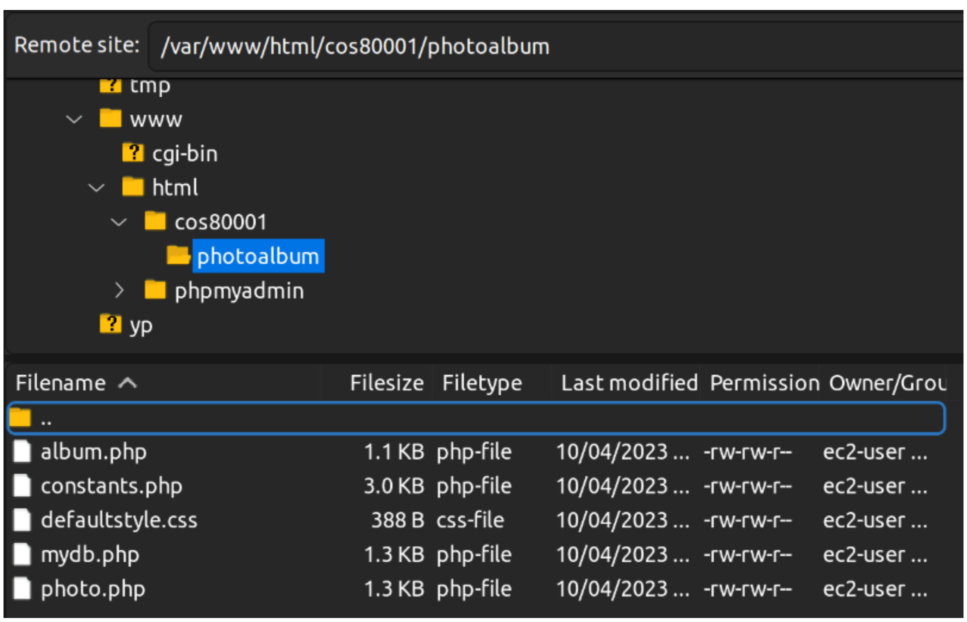
album.php page displayed from EC2 Web server

1. Change the constants.php file to your information




```php
*/

// [ACTION REQUIRED] your full name
define('STUDENT_NAME', 'Trac Duc Anh Luong');
// [ACTION REQUIRED] your Student ID
define('STUDENT_ID', '103488117');
// [ACTION REQUIRED] your tutorial session
define('TUTORIAL_SESSION', 'Saturday 10:00 AM');

// [ACTION REQUIRED] name of the S3 bucket that stores images
define('BUCKET_NAME', '103488117-assignment1b');
// [ACTION REQUIRED] region of the above bucket
define('REGION', 'us-east-1');
// no need to update this const
define('S3_BASE_URL','https://'.BUCKET_NAME.'.s3.amazonaws.com/');

// [ACTION REQUIRED] name of the database that stores photo meta-data (note that this is not the DB identifier of the RDS in
define('DB_NAME', 'assignment1b');
// [ACTION REQUIRED] endpoint of RDS instance
define('DB_ENDPOINT', 'assignment1b-testdb.ckti4ybwotgj.us-east-1.rds.amazonaws.com');
// [ACTION REQUIRED] username of your RDS instance
define('DB_USERNAME', 'admin');
// [ACTION REQUIRED] password of your RDS instance
define('DB_PWD', 'ducanh2003');

// [ACTION REQUIRED] name of the DB table that stores photo's meta-data
define('DB_PHOTO_TABLE_NAME', 'photos');
// The table above has 5 columns:
// [ACTION REQUIRED] name of the column in the above table that stores photo's titles
define('DB_PHOTO_TITLE_COL_NAME', 'photo_title');
// [ACTION REQUIRED] name of the column in the above table that stores photo's descriptions
define('DB_PHOTO_DESCRIPTION_COL_NAME', 'description');
// [ACTION REQUIRED] name of the column in the above table that stores photo's creation dates
define('DB_PHOTO_CREATIONDATE_COL_NAME', 'creation_date');
// [ACTION REQUIRED] name of the column in the above table that stores photo's keywords
define('DB_PHOTO_KEYWORDS_COL_NAME', 'keywords');
// [ACTION REQUIRED] name of the column in the above table that stores photo's links in S3
define('DB_PHOTO_S3REFERENCE_COL_NAME', 's3_reference');
?>
```

2. Transfer the files to the Ec2 Bastion Instance



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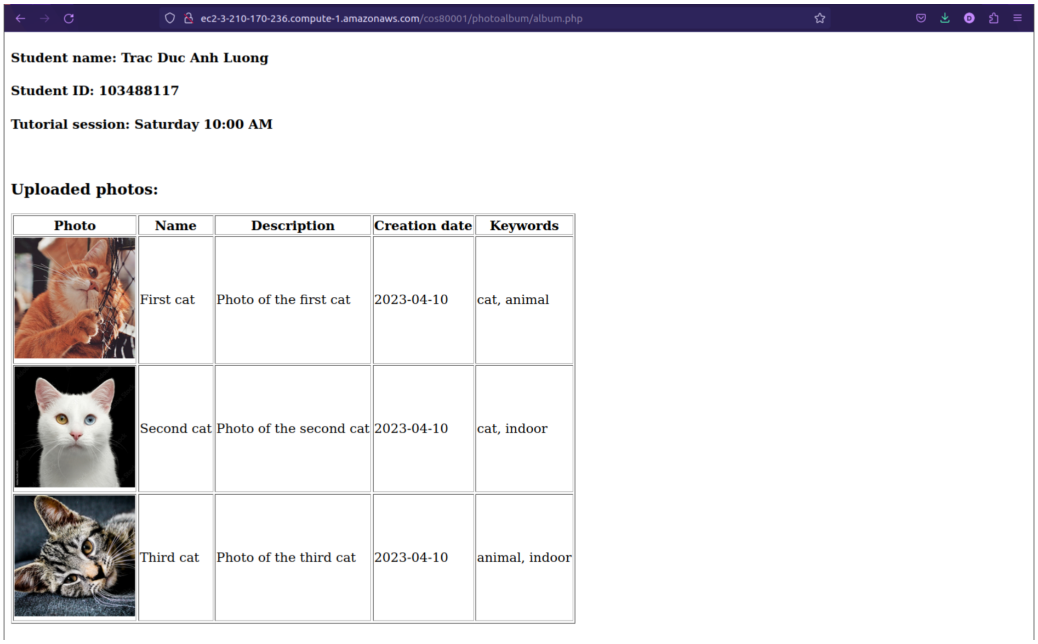
3. album.php page displayed from EC2 Bastion Instance


ec2-3-210-170-236.compute-1.amazonaws.com/cos80001/photoalbum/album.php

Student name: Trac Duc Anh Luong

Student ID: 103488117

Tutorial session: Saturday 10:00 AM

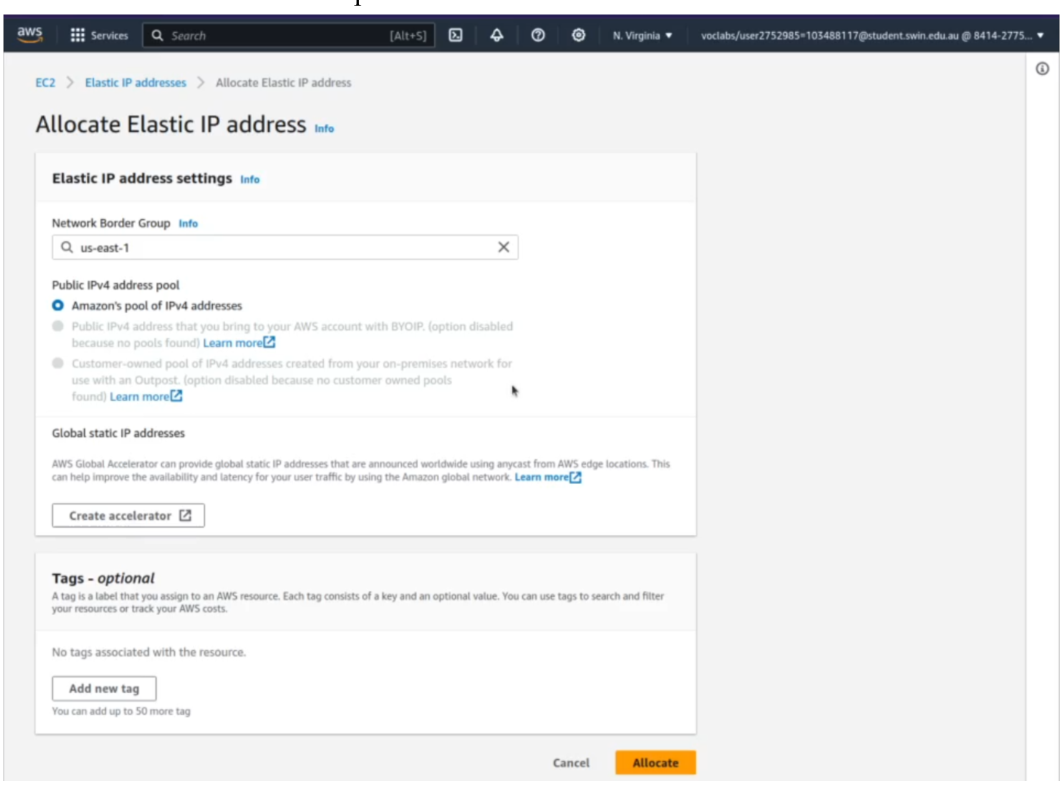
Uploaded photos:

Photo	Name	Description	Creation date	Keywords
	First cat	Photo of the first cat	2023-04-10	cat, animal
	Second cat	Photo of the second cat	2023-04-10	cat, indoor
	Third cat	Photo of the third cat	2023-04-10	animal, indoor

Provided URL is persistent (Elastic IP Association)

1. Allocate Elastic Ip address

- Network Border Group: us-east-1



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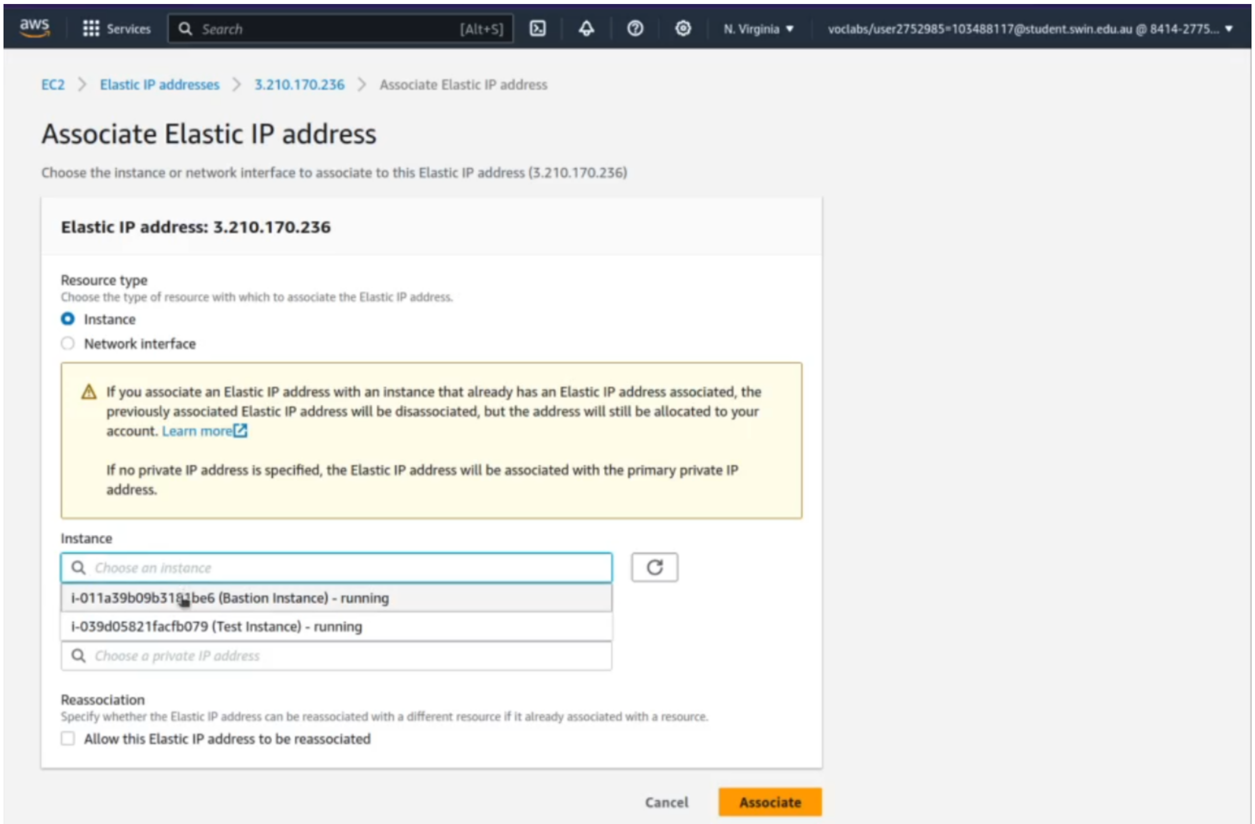
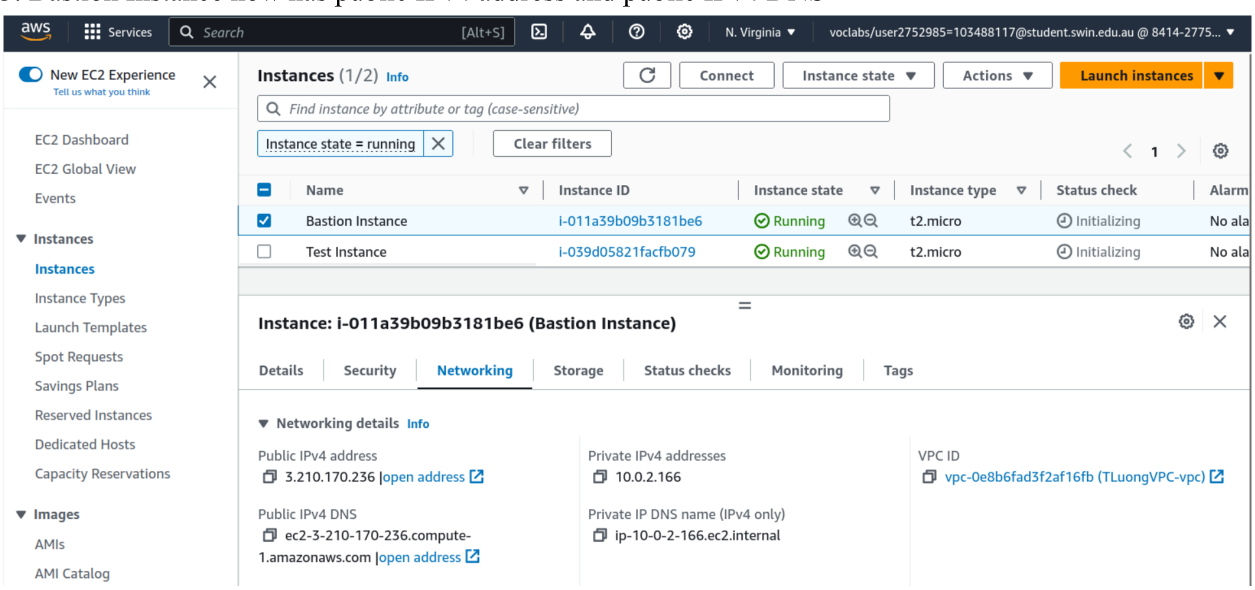
2. Associate Elastic IP address
   - Instance: Bastion Instance



3. Bastion Instance now has public IPv4 address and public IPv4 DNS



4. Public IPv4 address: 3.210.170.236
5. Website URL: ec2-3-210-170-236.compute-1.amazonaws.com/cos80001/photoalbum/album.php

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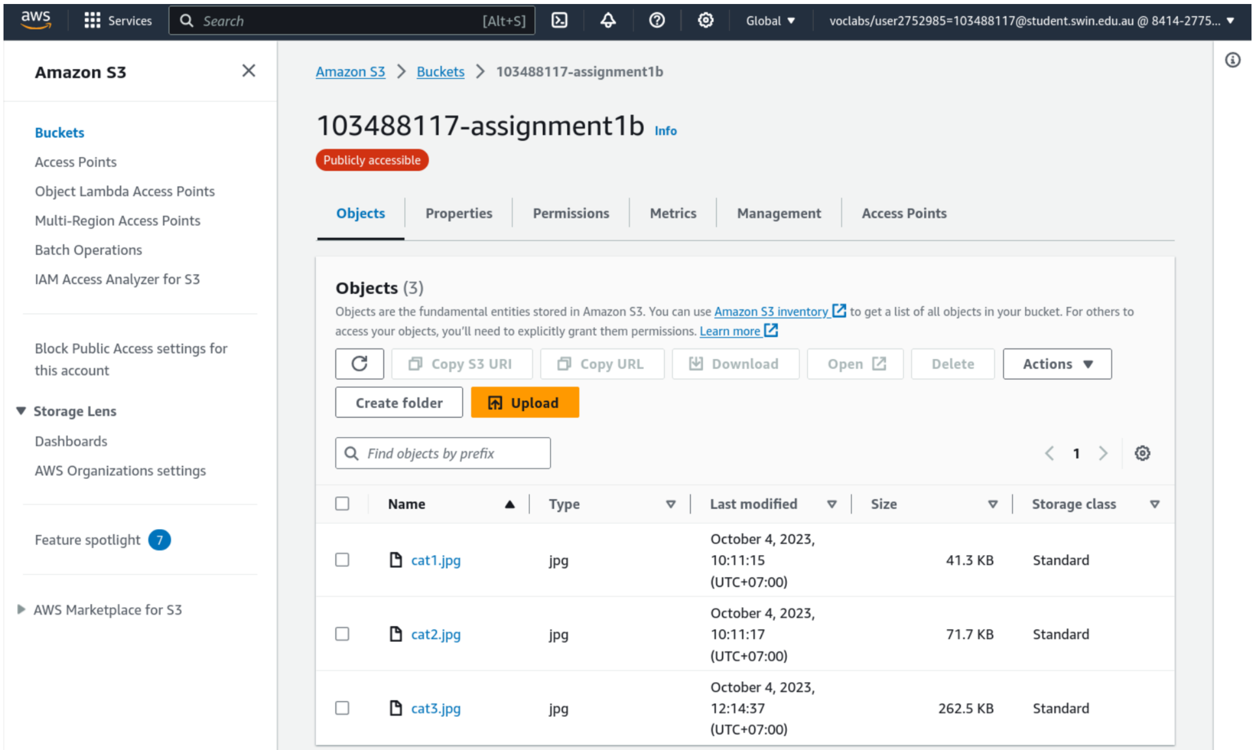
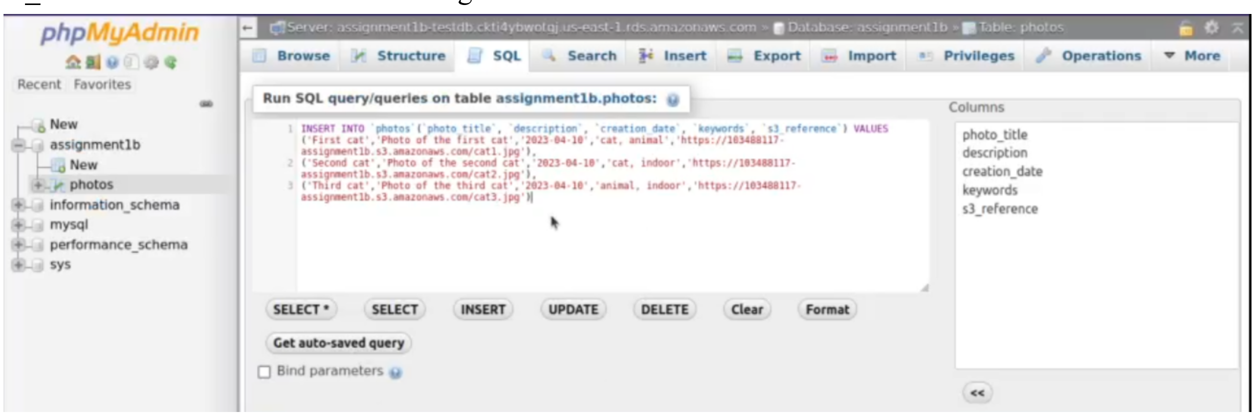
Photos loaded from S3 with matching metadata from RDS

1. Upload photos to S3



2. Insert SQL statements to the database table
s3_reference will be the link to the images in the S3 bucket.



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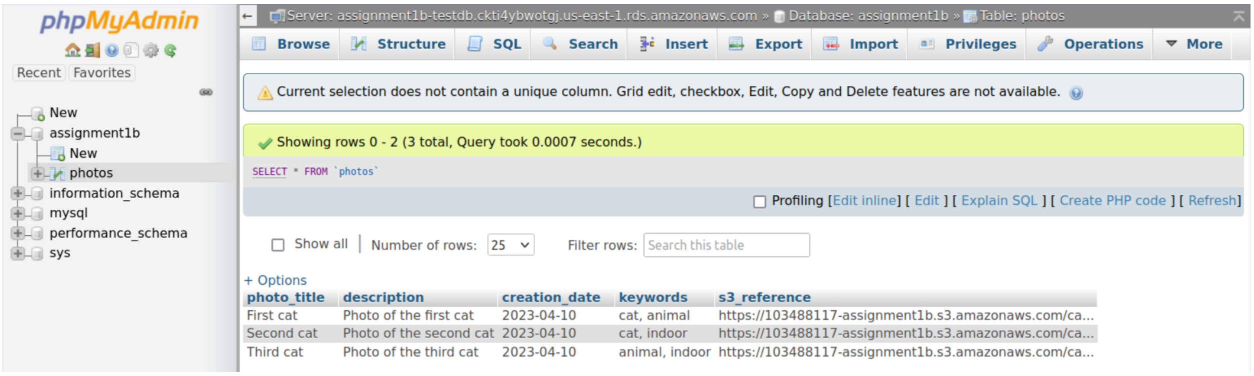
3. Database records



photo_title	description	creation_date	keywords	s3_reference
First cat	Photo of the first cat	2023-04-10	cat, animal	https://103488117-assignment1b.s3.amazonaws.com/ca...
Second cat	Photo of the second cat	2023-04-10	cat, indoor	https://103488117-assignment1b.s3.amazonaws.com/ca...
Third cat	Photo of the third cat	2023-04-10	animal, indoor	https://103488117-assignment1b.s3.amazonaws.com/ca...

4. Photos loaded from S3 with matching metadata from RDS

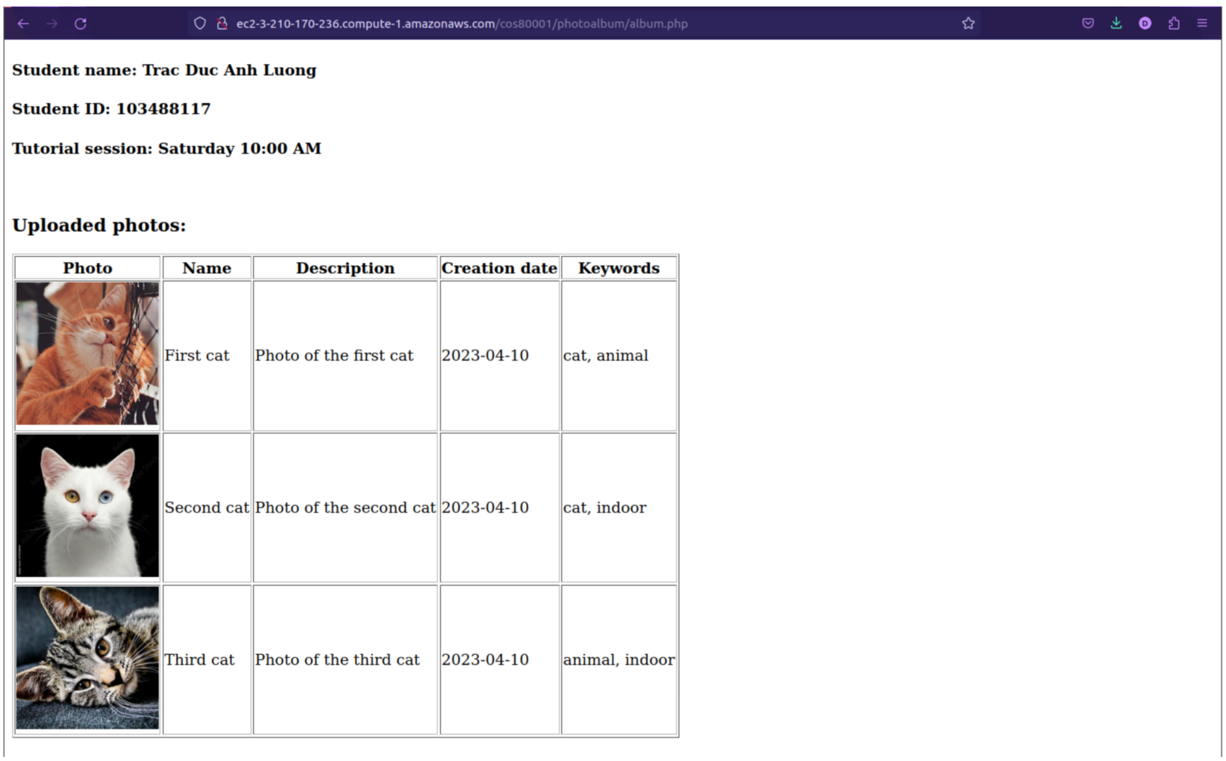


Photo	Name	Description	Creation date	Keywords
	First cat	Photo of the first cat	2023-04-10	cat, animal
	Second cat	Photo of the second cat	2023-04-10	cat, indoor
	Third cat	Photo of the third cat	2023-04-10	animal, indoor

Name: Trac Duc Anh Luong - ID: 103488117

Web server instance reachable from Test instance via ICMP

1. Ssh to Bastion Instance (10.0.2.166)

```
 ~/Desktop/COS20019 ------------------------------------------------------------
> ssh -i LTDA-linux.pem ec2-user@ec2-3-210-170-236.compute-1.amazonaws.com
Last login: Wed Oct  4 04:48:47 2023 from 116.96.47.180
   ,     #_
   ~\_  ####_        Amazon Linux 2
  ~~  \_#####\
  ~~     \###|       AL2 End of Life is 2025-06-30.
  ~~       \#/ ___
   ~~       V~' '->
    ~~~         /    A newer version of Amazon Linux is available!
      ~~._.   _/
         _/ _/       Amazon Linux 2023, GA and supported until 2028-03-15.
       _/m/'           https://aws.amazon.com/linux/amazon-linux-2023/
```

2. Ssh to Test Instance (10.0.2.210) from Bastion Instance

```
[ec2-user@ip-10-0-2-166 ~]$ ssh -i LTDA-linux.pem ec2-user@10.0.4.210
Last login: Wed Oct  4 04:58:45 2023 from ip-10-0-2-166.ec2.internal
   ,     #_
   ~\_  ####_        Amazon Linux 2
  ~~  \_#####\
  ~~     \###|       AL2 End of Life is 2025-06-30.
  ~~       \#/ ___
   ~~       V~' '->
    ~~~         /    A newer version of Amazon Linux is available!
      ~~._.   _/
         _/ _/       Amazon Linux 2023, GA and supported until 2028-03-15.
       _/m/'           https://aws.amazon.com/linux/amazon-linux-2023/
```

3. From Test Instance, ping Bastion Instance and provide the result

```
[ec2-user@ip-10-0-4-210 ~]$ ping 10.0.2.166
PING 10.0.2.166 (10.0.2.166) 56(84) bytes of data.
64 bytes from 10.0.2.166: icmp_seq=1 ttl=255 time=0.434 ms
64 bytes from 10.0.2.166: icmp_seq=2 ttl=255 time=0.567 ms
64 bytes from 10.0.2.166: icmp_seq=3 ttl=255 time=0.546 ms
64 bytes from 10.0.2.166: icmp_seq=4 ttl=255 time=0.551 ms
64 bytes from 10.0.2.166: icmp_seq=5 ttl=255 time=0.657 ms
64 bytes from 10.0.2.166: icmp_seq=6 ttl=255 time=0.583 ms
64 bytes from 10.0.2.166: icmp_seq=7 ttl=255 time=0.603 ms
64 bytes from 10.0.2.166: icmp_seq=8 ttl=255 time=0.532 ms
64 bytes from 10.0.2.166: icmp_seq=9 ttl=255 time=0.542 ms
64 bytes from 10.0.2.166: icmp_seq=10 ttl=255 time=0.553 ms
^C
--- 10.0.2.166 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9210ms
rtt min/avg/max/mdev = 0.434/0.556/0.657/0.061 ms
[ec2-user@ip-10-0-4-210 ~]$
```